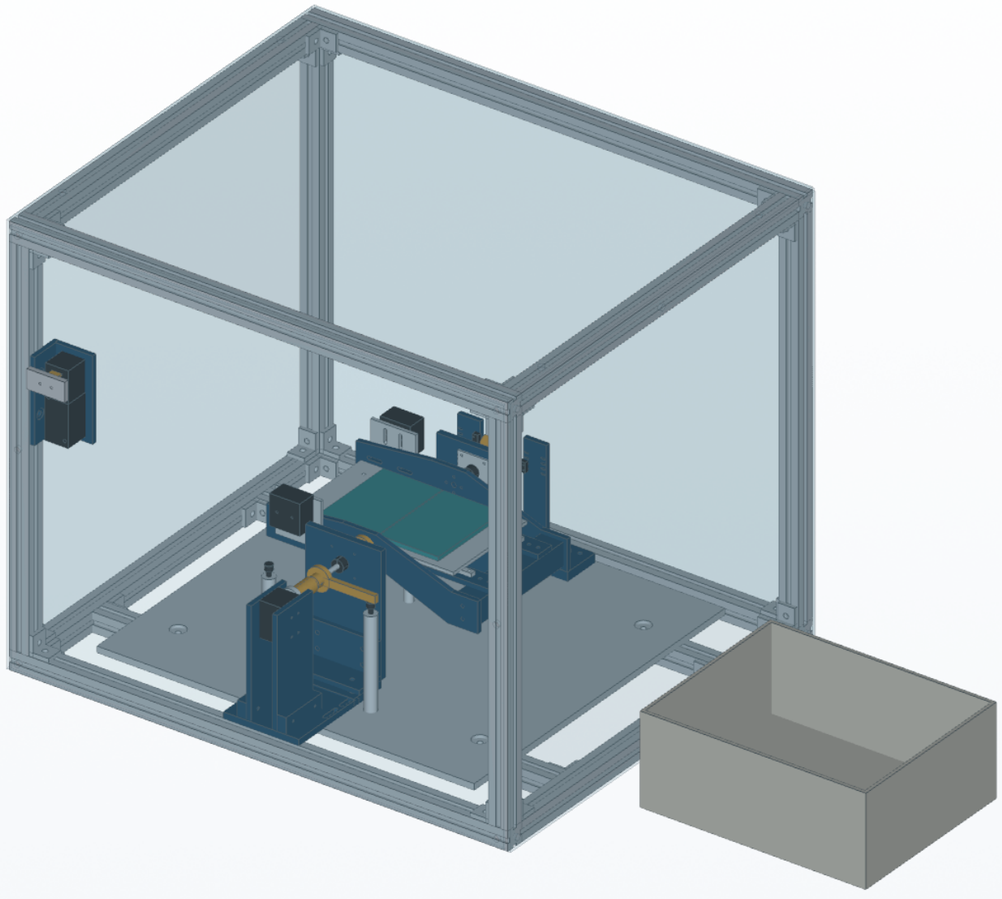
The bench, resolved in CAD

1010 one-inch extrusion | vendor interfaces | machined fixture | guarded development station



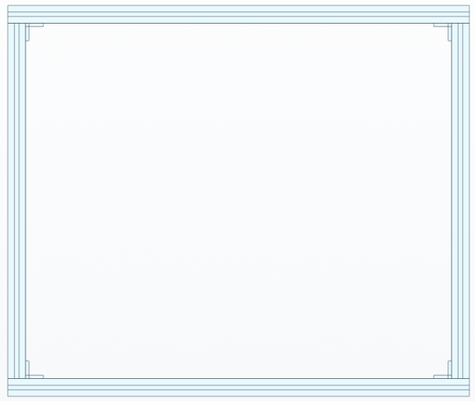
A real geometry model
OpenCascade solids, STEP assembly, individual custom STEP parts and millimetre DXF profiles. The views in this book are generated from that same model.

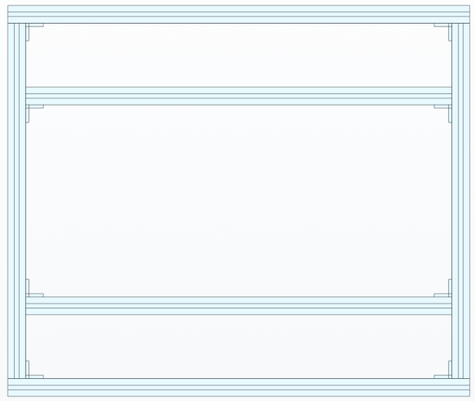
A traceable source package
Manufacturer drawings are indexed by source ID. Original vendor files are separated from reconstructed interface models and custom designs.

Build status
This issue supports a development-shop review and prototype quotation. It is not a fully released build package. P1-P4 identify the interfaces that still prevent complete assembly and powered commissioning.

Datums and installation envelope

A = top of fixture base | X = across phone | Y = hinge axis | Z = up

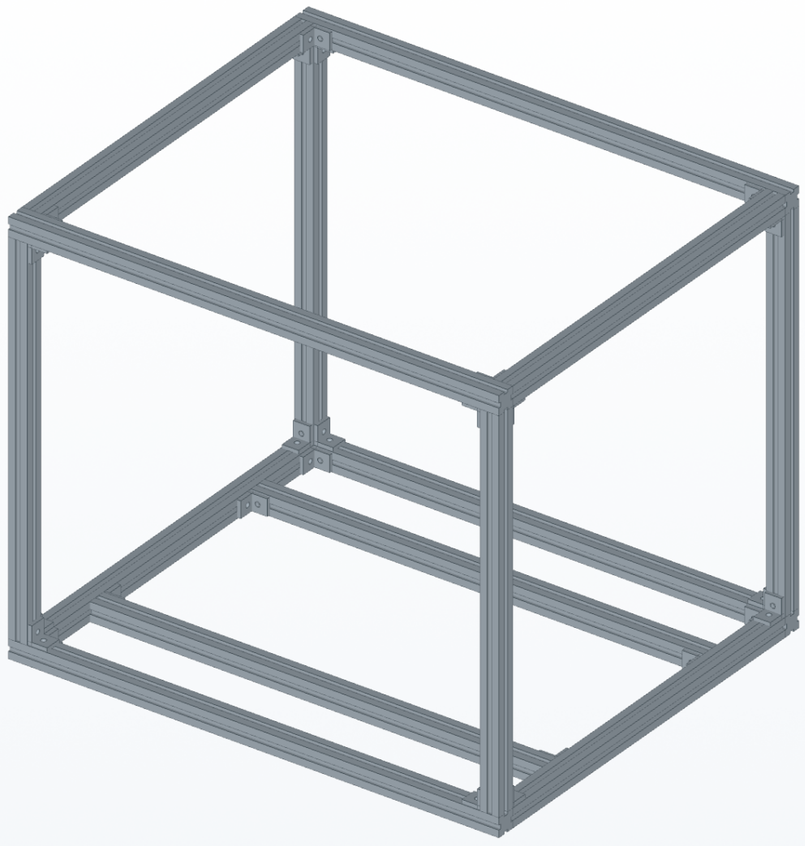




Dimension / location	Value	Basis
Frame outside, excluding panel thickness	660.4 x 558.8 x 558.8	26 x 22 x 22 in; 1010 is 25.4 mm square
Frame underside / roof extrusion top	Z=-33.4 / +525.4	Base top is zero; guard roof top +529.4
Fixture base / support crossmembers	500 x 460 x 8 / Y=+/-150	Base top Z0; frame rail top Z=-8
Hinge axis / bearing tower outside faces	X0, Z140 / Y=+/-124	Bearing pockets face inward; fixture bolts allow alignment
Cabinet envelope; independent on bench	299.33 x 239.46 x 119.73	Hammond 1554YA2GY, source S13; no unsupported frame cantilever
Recommended work surface	1200 x 800 minimum	Project allowance for cabinet, leads and service access

Frame cut list and joint schedule

80/20 1010 / 4119 / 3393 | do not substitute metric 2020 hardware



Member	Qty	Cut length	Location
V01	4	660.4 +/-0.25	Front/rear; top/bottom
V02	4	508.0 +/-0.25	Side rails; top/bottom
V02	4	508.0 +/-0.25	Vertical posts
V03	2	609.6 +/-0.25	Base crossmembers

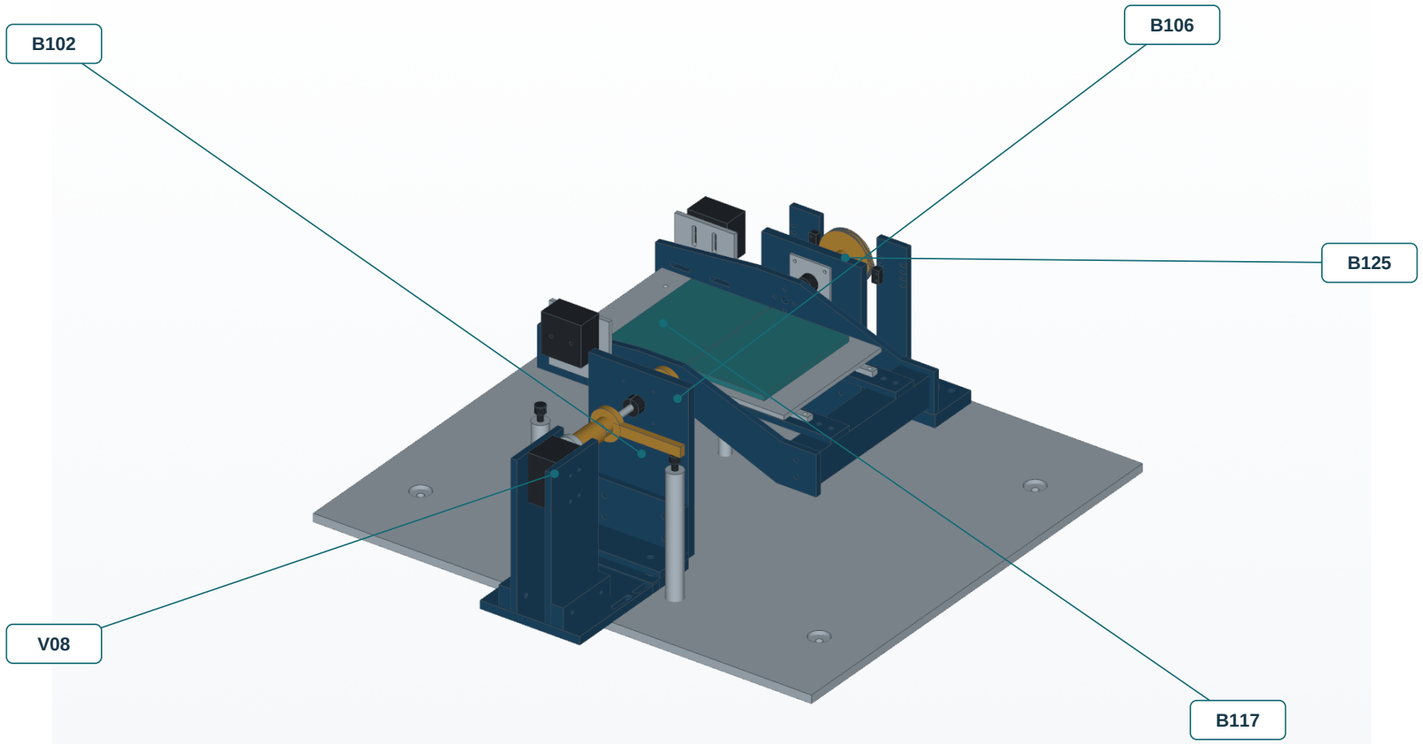
Total cut length: 7,924.8 mm. Order cut-to-length or add saw kerf and end-trim waste to stock. Match paired members within 0.10 mm; deburr ends.

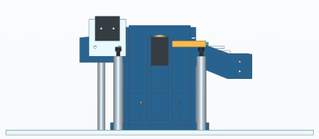
Joints	4119	3393 sets
8 horizontal corner joints	8	16
4 posts x 2 ends x 2 directions	16	32
2 crossmembers x 2 ends	4	8
Total structural joints	28	56

The model includes all 28 brackets. The schedule above governs their hardware quantities. Confirm wrench access and bracket-to-bracket clearance at the first corner before repeating. Frame diagonal difference <=0.5 mm per rectangle; no foot rocking on the installation surface.

Fixture arrangement and motion

Counterbalanced yoke | twin supported stub shafts | radial float | physical output stop cam





Fixed half: B117 on four B118 posts. B117 top is Z136.4; 1 mm pad gives nominal contact surface Z137.4.

Moving half: B109 plate on two MGN7C blocks; B108 carriers bolt to B107. Axis is X0/Z140; yoke rotates about Y. The phone envelope is unverified, not an Apple model.

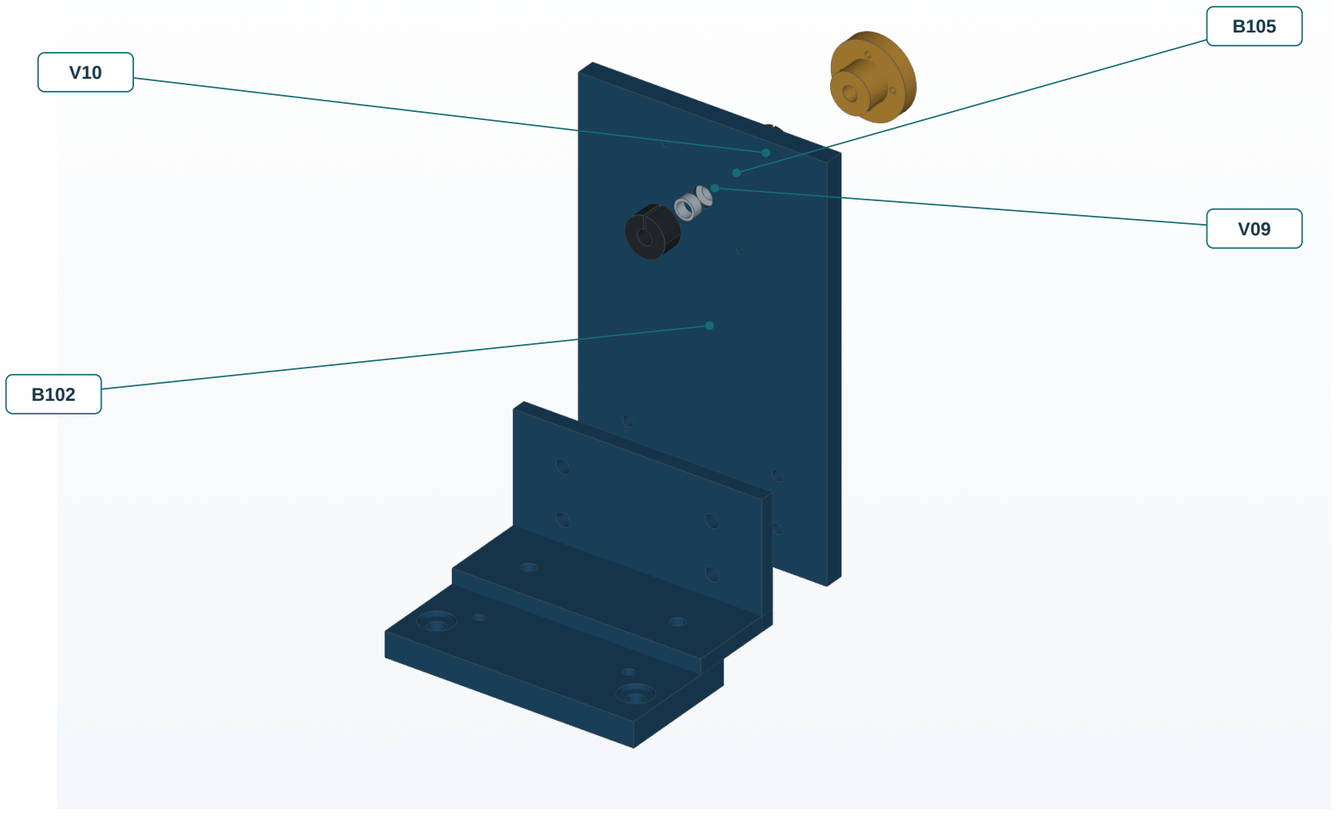
Float: block centers X55; four B110 stops produce 5.7..6.0 mm travel each way after 2 mm pads are fitted, depending on end-seal length. Rails start X15 and finish X115.

Balance: move both B120 plates in X, then B111 weights in Z. Measure residual gravity torque with the actual dummy and all wiring installed. Goal <=0.02 N m; do not substitute the CAD mass estimate for this measurement.

Stops: B123 is on the output shaft, so the slip clutch cannot hide output overtravel. B124 posts carry adjustable M6 screws and soft pads. Normal operation must stop short of these mechanical stops.

Bearing and shaft stack

Use inner-race spacers; do not preload bearings by tightening collars



Press only the bearing outer race into the tower seat. Never transmit press force through the balls.

Retainer B105 has D17 opening. Pocket depth 6.05 leaves at least 0.05 mm outer-race clearance.

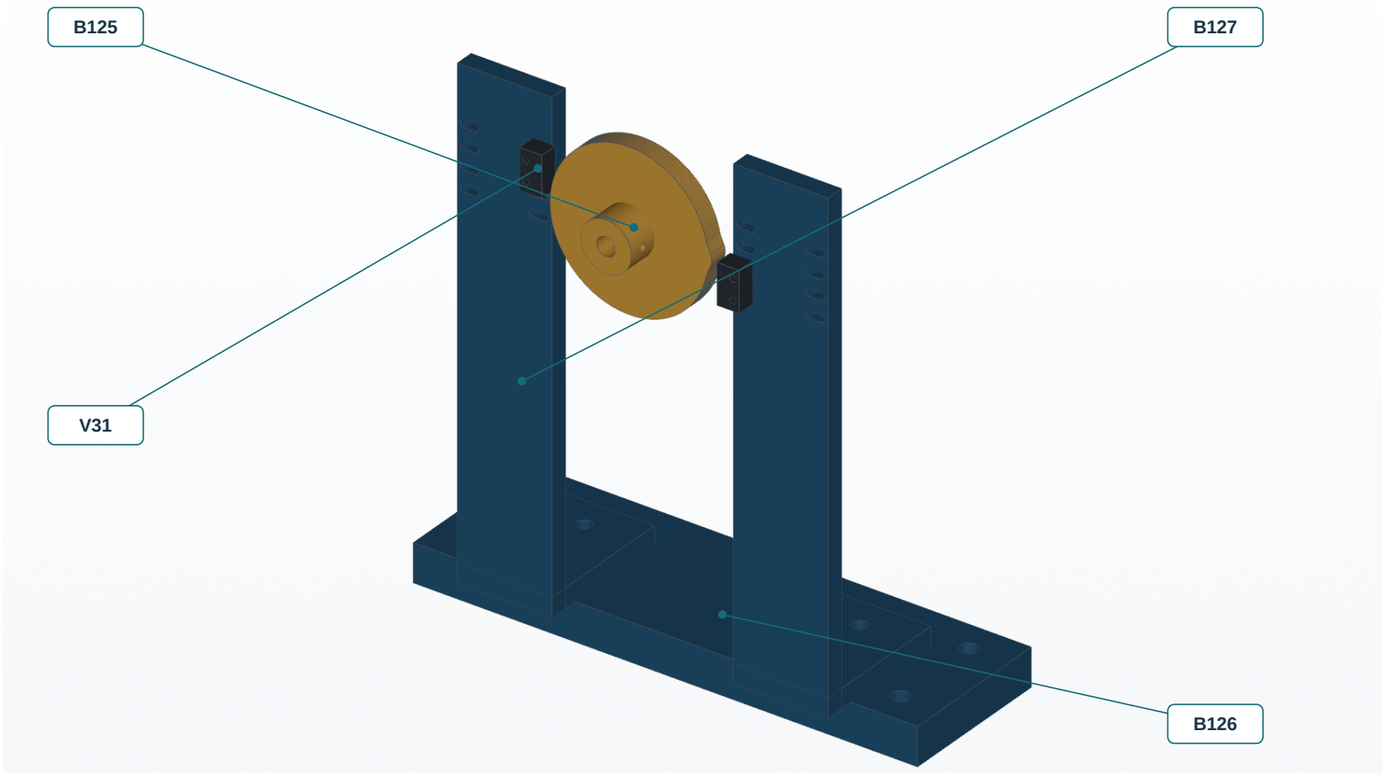
Set collars with 0.05-0.10 mm total axial freedom. Ruland MCL-6-A screw torque is 2.1 N m (S08).

Flat clocking and clutch engagement remain P2. Do not cut the output shaft short to suit an assumed clutch internal dimension.

Front-side item	Y interval / position	Interface
B113 output shaft	-193 to -85	D6 g6, 108 long
B123 cam	-180 to -164	D6 H7 bore; two flats at Y=-175
Outer collar / B116 spacer	-135.05..-126.05 / -126.05..-122.05	Collar D16; spacer D8 clears tower D8.2
NSK 626 bearing	-122.05..-116.05	6 x 19 x 6; pocket D19 H7
B115 spacer / inner collar	-116.05..-114.05 / -114.05..-105.05	Spacer contacts inner race only
B112 hub / B106 arm	-101..-85 / -85..-79	4 x M3 on PCD20; two shaft flats at Y=-96
Rear stack	Mirror bearing/hub stack; shaft +85..+169	B114 = 84 long; B125 endpoint cam on rear tail

Output-side endpoint sensing

B125 sensing cam / B126 sole / two B127 uprights | normal position feedback, not a safety stop



Mounting datums
Cam axis X0,Z140; barrel Y151..161, discY161..167. Sole center X0,Y202, bottomZ0. Upright origins X+/-43.8,Y167,Z12.

Exact switch
OMRON D2F-01L-D3 has solder terminals. Both mounting-hole pairs are6.5mm apart. Right body center Z133.6; left Z146.4, rotated180deg about Y. This places both lever tips approximately at Z140. See S24.

Radial adjustment
Start hole baselines at X+/-32.8. Adjust each switch in its6x2.4 slots with a continuity meter. Mounting torque0.08..0.10Nm; do not crush the switch body.

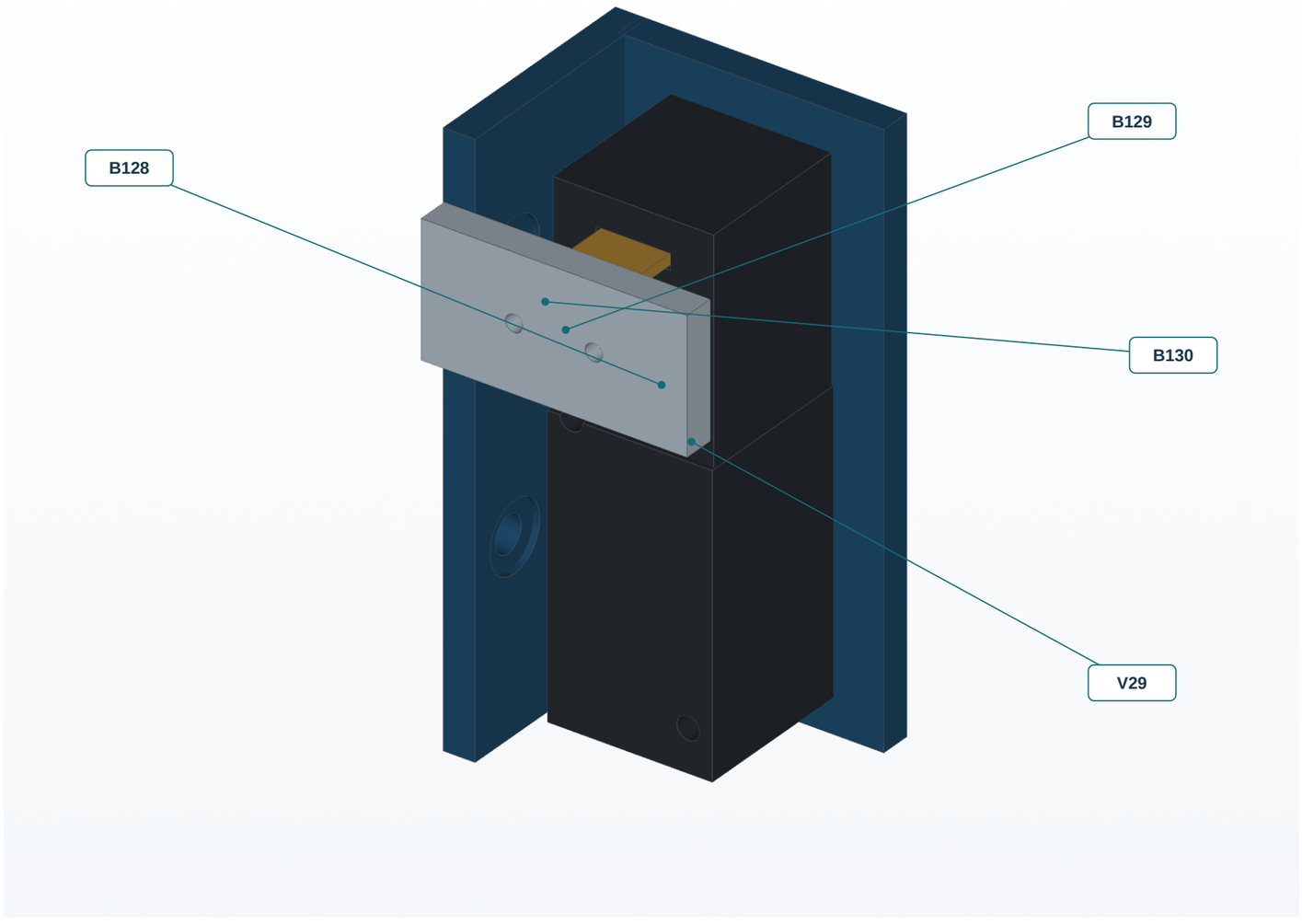
Travel acceptance
Measure electrical make, release and actual remaining lever travel. At the cam peak, target0.35..0.45mm beyond make, within the published minimum available overtravel0.55mm. Test the whole sweep; a nominal OP dimension cannot establish a safe setting.

Timing
Set signals near fixture5deg and175deg, before backup stops0/180deg. Confirm one signal at each end and both low through mid-travel. Inspect cam retention and wear during dummy endurance testing.

Element	Verified source / design boundary
D2F-01L-D3	OP6.8+/-1.5; MDmax0.5; OTmin0.55; hinge lever maximum force0.78N (S24)
B125	Project cam: R25 base, R26.2 peak, cosine rise over+/-12deg. Actual make/release measured.
CAD image	Switch bodies are interface envelopes; spring-lever deflection and solder tails are not modeled.
P3	Endpoint hardware is dimensioned. Physical actuation, insertion and access tests remain open.

Guard interlock / reinforced key mount

OMRON D4NS-1BF + D4DS-K2 | B128-B130 | straight withdrawal of screw-removable front panel



Frame mount
B128 origin(-275,-238.15,255). Its left face contacts the left-front post at X=-304.8. Preload two3393 T-nuts in that post's inward-facing slot. Recesses preserve the vendor bracket's4.775mm clamp stack.

Switch mount
Rear seating planeY=-238.15. Two diagonal body bolts: (-286,316.5) and(-264,269.5) in global X,Z. Set head to front insertion. Cut M4x50 blanks to measured stack plus nut and two threads; record length.

Key mount
Key hole centers X=-282.5/-267.5,Z344. B129 backs the outside of the front panel. B130 sleeves carry the clamping force through the D7 panel holes. Measure panel thickness before finishing sleeve length.

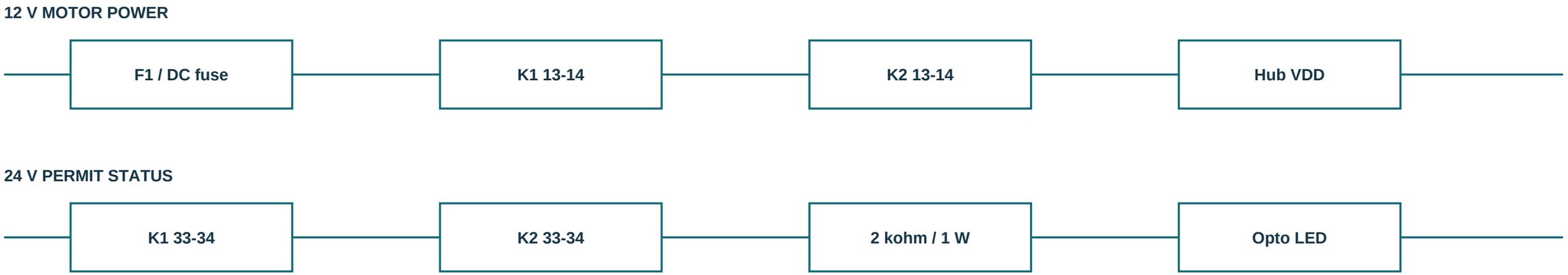
Insertion datum
Nominal rear-plane to key seating-plane distance41.15mm with4.1mm sleeves. Actual must be40..42.5mm. Align key tip to opening within+/-1mm in both directions; key-slot center is nominalZ350.

Physical acceptance P3
Guard seats against frame, not against the switch. Withdraw squarely without levering the key. Test both channels and loss of permit before reachable motion. Validate stopping distance and anti-tamper measures; a fitted key alone does not establish guard safety.

Connection	Required assembly check
Switch body	0.5..0.7Nm; no distorted housing. Use actual body seating/bolt recesses, not envelope depth.
Operation key	2.4..2.8Nm. Verify sleeve ends bear on metal; plastic must not carry the bolt preload.
Drawing scope	Switch/key are reconstructed interface envelopes, not original OMRON manufacturing CAD.
Cable entry	D4NS-1BF is PG13.5. Use compatible plastic gland; cabinet M16 gland does not fit this entry.

Independent stop circuits / wiring review

Exact PNOZ X3 terminal references | two separate input devices | HOLD P4 before energizing

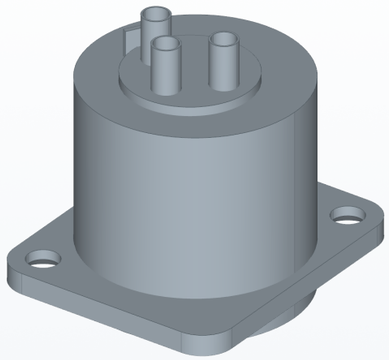


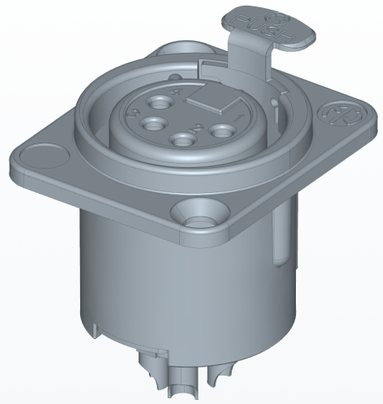
Circuit	Connection	Rule / source
Both relays: supply	+24V -> B1; 0V24 -> B2	774310 DC variant; A1/A2 unused
K1 E-stop channel A	S31 -> E-stop NC block A11 -> A12 -> S32	Direct-opening NC; source S20 p12
K1 E-stop channel B	S21 -> E-stop NC block B11 -> B12 -> S22	Two distinct contact blocks; do not join channels
K2 guard channel A	S31 -> D4NS11 -> 12 -> S32	Use actual device terminal markings
K2 guard channel B	S21 -> D4NS21 -> 22 -> S22	No E-stop contacts in series in these inputs
Both relays: link	S11 -> S12	Cross-short-detection configuration, S20 p12
K1 and K2: separate reset	S33 -> respective NO reset -> S34	Monitored reset; S13/S14 remain unlinked; S20 p13
Unused outputs	23/24,41/42,Y31/Y32	Leave isolated; no undocumented bridges

Review limits: F1=V52(2A), F2=V53(1A). Fuse coordination, connector adapter cables, wiring protection and terminal allocations are not released by this diagram. Verify relay load/inrush ratings, supply fault current and the current manufacturer manual. The drawing does not assign a machine performance level or replace a stopping test.

Controller interface and connector details

Active-high signals preserved for existing FastAPI driver | no 24 V on Pi header





Original vendor CAD
These connector views are rendered from Neutrik's downloaded STEP files, not reconstructed cylinders. Original models are supplied in the vendor reference archive.

Panel pattern
D24.0 +0.1/0 opening; two D3.2 mount holes at (-9.5,+12) and (+9.5,-12), measured from connector center. Mirror only if the vendor drawing shows the opposite viewing face. Trial-fit a scrap panel before cutting the enclosure.

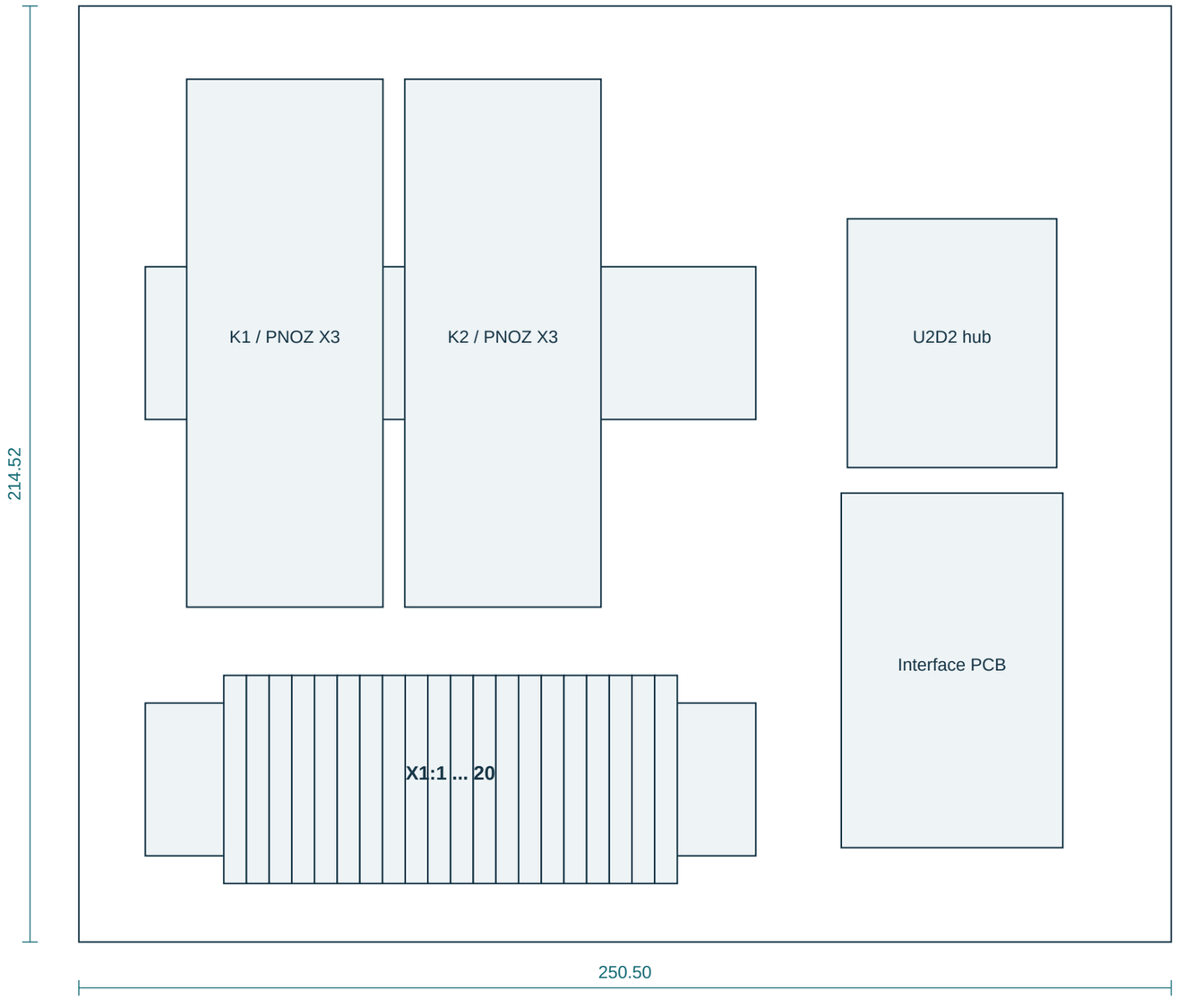
Project pinout
NC3: 1=0V12,2=+12V,3 unused.
NC4: 1=0V24,4=+24V,2/3 unused.
Label both DC ONLY. Inspect solder-cup pin numbers; do not wire from a front-view mirror image.

Signal	BCM / header	Connection / inspection
Safety permit	GPIO17 / pin11	LTV-817C collector to3.3V; emitter to node;10k node-to-GND;1k node-to-pin11. High=permit.
Open endpoint	GPIO27 / pin13	Switch COM to Pi5V(pin2). NO ->1.8k,1% ->GPIO27; 2.7k,1% GPIO27-to-GND. Nominal high3.0V, switch1.11mA.
Closed endpoint	GPIO22 / pin15	Same5V switch/1.8k+2.7k divider as open endpoint. Do not connect5V directly to a GPIO pin.
Temperature data	GPIO4 / pin7	Two powered DS18B20 sensors;one4.7k pullup to3.3V. Record both serial numbers.
Logic supply / return	pin1 / pin6	3.3V / Pi GND. Endpoint COM only uses5V pin2. No direct24V connection. Verify GPIO highs2.0..3.3V.
TTL motor lead	JST EHR-03	1=GND,2=VDD12V,3=DATA. Use U2D2 TTL port; not RS-485.

Opto input: K2:34 -> 2 kohm /1W -> LED anode; LED cathode ->0V24. Fit1N4148 antiparallel across the LED. Check the selected LTV-817C package pinout/CTR and cover the board. This status circuit is not the safety power-cut path. Endpoint correction: S24's microload graph starts at5V/1mA. The earlier3.3V/10k switch circuit is outside that graph. Require measured COM supply5.00..5.25V under full Pi load. With1% resistors, calculated GPIO high is2.976..3.175V and contact current is at least1.10mA. If supply is below5V, hold commissioning pending a qualified regulated supply. Measure with GPIO disconnected before attaching the Pi.

Cabinet backplate / component allocation

Hammond1554YPL | front view | datum at panel center | fit review before drilling



Item	Center X,Y	Envelope / allocation
Upper DIN rail	-40,+30	140 x35; 7.5 high
Lower DIN rail	-40,-70	140 x35; 7.5 high
K1 / K2	-78,+30 / -28,+30	45 x121 each;87 deep
X1 bank	-40,-70	20x5.2=104;47.7 tall
Hub PCB	+75,+30	48 x57; use supplied10mm supports
Interface PCB	+75,-45	50.8 x81.28; long axisY
Terminal end cover	Open side of bank	2.2 wide
End brackets	Either side of each bank	5.2 wide each

Defined fit: terminal bank + cover + two end brackets =116.6mm, within140mm rail. Relay bank including 5mm gap and two end brackets =105.4mm. Nominal gap between relay body and terminal body is15.65mm.

Depth check: catalogue enclosure internal height111.75mm; relay87mm + rail7.5mm leaves17.25mm before panel/support offsets and lid features. Measure the actual stack; this is not a verified terminal-wire or lid-button clearance.

Drilling: preserve the factory panel mounting holes. Locate rail fixings using actual slots; use at least two fixings per rail. Transfer the hub's purchased mounting pattern. Neither vendor pattern is invented here.

P4: E-009 locates operators/inlets. Fuse-holder restraints and remaining harness entries require fit review. Keep the Pi outside. No component on the lid may occupy the relay depth envelope.

Coordinates are a dimensioned allocation plan, not a released drill template. The panel's factory corners and holes are omitted in this view; preserve them.

Terminal allocation and bus inspection

X1 =20 Phoenix UT2,5 terminals | separate12V,24V and Pi logic

Terminal	Net	Side A	Side B / bridge
X1:1	12V inlet	NC3 pin2	F1 input
X1:2	0V12	NC3 pin1	Power-hub GND
X1:3	24V inlet	NC4 pin4	F2 input
X1:4	0V24 inlet	NC4 pin1	Wire to X1:9 sideA
X1:5	+24V fused	F2 output	K1:B1; FBS4-5 bridges5..8
X1:6	+24V fused	K2:B1	Spare; bus5..8
X1:7	+24V fused	K1:33 permit feed	Spare; bus5..8
X1:8	+24V fused	Spare	Spare; bus5..8
X1:9	0V24	Wire from X1:4 sideB	K1:B2; bridge9..12
X1:10	0V24	K2:B2	Spare; bus9..12
X1:11	0V24	Opto LED cathode	Spare; bus9..12
X1:12	0V24	Spare	Spare; bus9..12
X1:13	Switched12V	K2:14	Power-hub VDD
X1:14..20	Reserved	Unconnected	No bridges installed

Direct safety-power wires: F1 output -> K1:13; K1:14 -> K2:13. The motor return is X1:2; do not switch it instead of positive power. Keep supply and motor wiring physically distinct from the Pi logic harness.

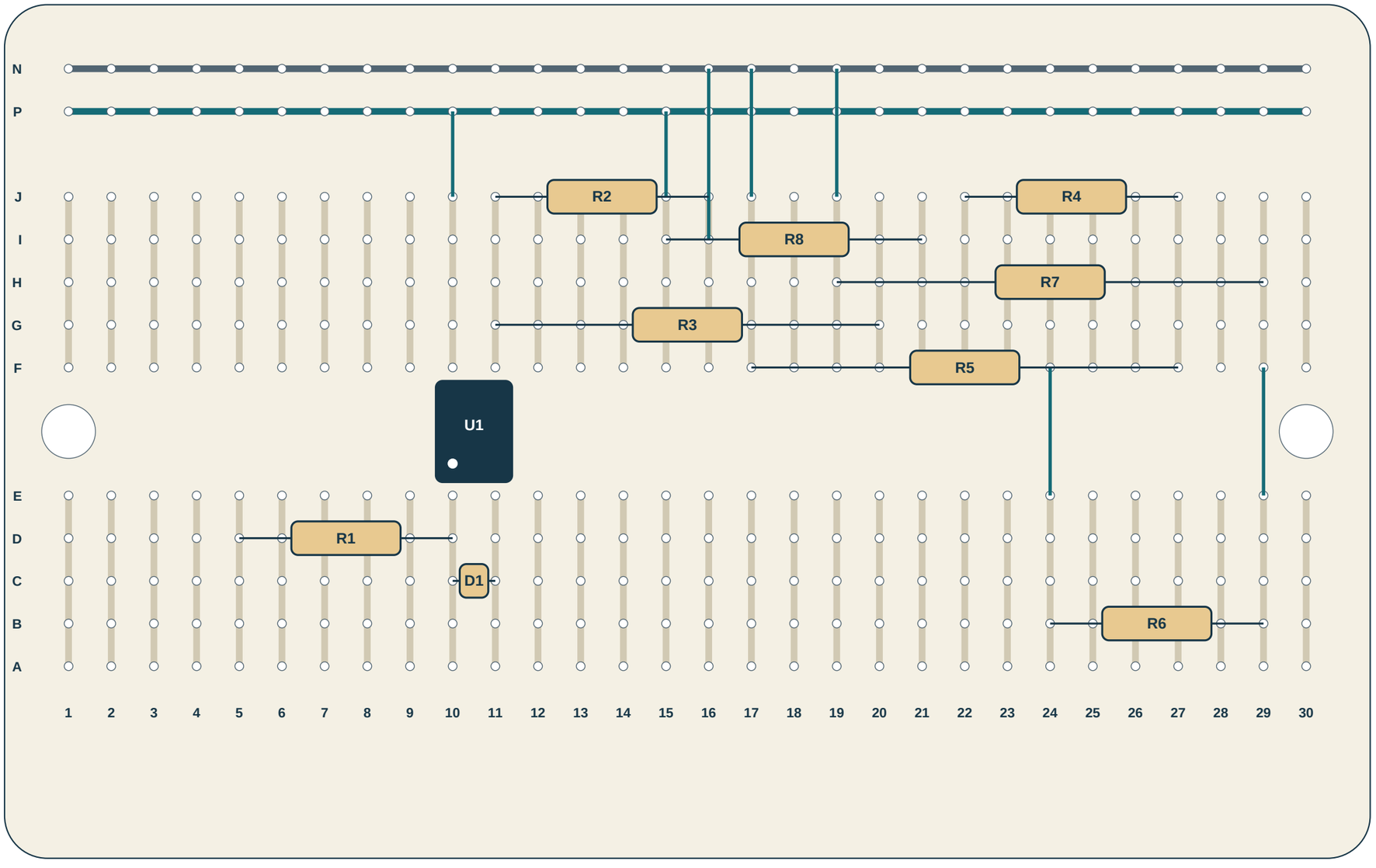
Permit chain: X1:7 -> K1:33; K1:34 -> K2:33; K2:34 ->2kohm/1W -> opto LED anode. Cathode ->X1:11. Use E-001 for channel and reset wiring; E-002 for GPIO wiring.

Assembly inspection: one conductor per clamping point. Strip9mm and tighten UT terminals0.5..0.6Nm. Number both ends of every wire. Before fitting bridges, continuity-check each terminal against adjacent terminals; after fitting, verify only5..8 and9..12 are common. No 12V/24V/Pi-GND cross-bridge.

P4 remains: wire sizes, cable jacket specifications, fuse coordination, exact termination accessories, cut lengths and physical routing require the final wiring release. This allocation does not authorize energizing the station.

Controller board / component-side placement

E-BOARD-01 | Adafruit1609 | dimensions mm | solder on reverse side



Read the real board
A-E are one copper strip per column; F-J are another. P is the upper inner rail atY=19.05 (+3.3V). N is the upper outer rail atY=21.59 (Pi GND). The two lower rails are unused and omitted here.

Orientation
View from component side, column1 left. U1 pins:1=E10,2=E11,3=F11,4=F10. White dot marks pin1. Do not mirror this view while soldering underneath.

Mounting
Two D3.2 holes on73.66 pitch; use insulating M3 supports. Keep metal and solder tails clear of the panel. Cabinet mounting and cover acceptance are P4.

Lead forming
Use insulated jumpers. Sleeve any exposed lead that crosses a pad or another lead. D1 is upright with an insulated return bend, cathode atC10.

Board grid derived from Limor Fried/Ladyada, Adafruit Perma-Proto original EAGLE CAD (S36). This derived placement drawing is CC BY-SA3.0; see THIRD-PARTY-NOTICES.md. Component bodies are placement symbols; use E-006 hole IDs.

Controller board / soldering schedule

One lead per listed hole | resistor bodies PR01 6.5 x2.5 nominal | do not substitute a solderless board

Ref	Component	Hole assignments
U1	LITEON LTV-817-C DIP4	E10, E11, F11, F10
R1	2k / 1% / 1W	D5, D10
D1	Vishay 1N4148; cathode first	C10, C11
R2	10k / 1% / 1W	J11, J16
R3	1k / 1% / 1W	G11, G20
R4	1.8k / 1% / 1W	J22, J27
R5	2.7k / 1% / 1W	F27, F17
R6	1.8k / 1% / 1W	B24, B29
R7	2.7k / 1% / 1W	H29, H19
R8	4.7k / 1% / 1W	I21, I15
W1	Insulated 24AWG jumper	I16 to N16
W2	Insulated 24AWG jumper	J10 to P10
W3	Insulated 24AWG jumper	J17 to N17
W4	Insulated 24AWG jumper	J19 to N19
W5	Insulated 24AWG jumper	J15 to P15
W6	Insulated 24AWG jumper	F24 to E24
W7	Insulated 24AWG jumper	F29 to E29

1 / Prepare: confirm PCB identity, clean pads and inspect for shipping damage. Verify A-E and F-J column connectivity and four separate rails with an unpowered meter. Compare the received PCB with the drawing before adding components.

2 / Fit: install insulated jumpers, resistors, D1 then U1. Form axial resistor leads outside their coated bodies. Raise bodies about2mm above PCB; keep all formed leads insulated at crossovers. Route external wires from the underside with insulated tails and strain relief.

3 / Inspect: solder using component-maker temperature limits. Trim tails to1mm maximum, clean flux, inspect under magnification for bridges, cold joints and lead damage. No solder bridge may replace an insulated jumper.

4 / Protect: board requires a screwed insulating cover and strain relief before cabinet acceptance. Keep24V wires at columns5..11 separate from the Pi harness. Neither this board nor U1 performs the motor safety-disconnect function.

Controller board / external terminations

Tag both ends | powered DS18B20 mode | header numbers are physical pin numbers

Board hole	Wire destination	Net
A5	K2:34 permit +24V	PERMIT24
A11	X1:11A 0V24	RETURN24
P30	Pi physical pin1 / 3.3V	3V3
N30	Pi physical pin6 / GND	GND
A25	Pi physical pin2 / 5V	5V
B25	Open switch COM	5V
C25	Closed switch COM	5V
I22	Open switch NO	OPEN_NO
J24	Closed switch NO	CLOSED_NO
J20	Pi physical pin11 / BCM17 permit	PERMIT_GPIO
I27	Pi physical pin13 / BCM27 open	OPEN_GPIO
J29	Pi physical pin15 / BCM22 closed	CLOSED_GPIO
F21	Pi physical pin7 / BCM4 data	ONEWIRE
G21	Case probe data	ONEWIRE
H21	Ambient probe data	ONEWIRE
P24	Case probe VDD	3V3
P25	Ambient probe VDD	3V3
N24	Case probe GND	GND
N25	Ambient probe GND	GND

Use 24AWG stranded insulated signal wire, minimum60V/80C. Use red for3.3V, orange for5V, black for Pi GND, violet for signals; label every conductor because color is not proof of voltage. Keep permit24V pair separately sleeved and labeled. Do not connect Pi GND to0V24. Verify the actual header pin1 mark before connecting. The final locking header harness and probe termination remain P4.

Controller board / inspection and bench test

Pi, motor and probes disconnected until low-voltage checks pass | meter readings become as-built evidence

Net	Holes that must be common before components
PERMIT24	A5, D5
RETURN24	A11, E11, C11
LED_ANODE	D10, E10, C10
3V3	F10, I15, P30, P24, P25
GND	J16, F17, H19, N30, N24, N25
EMITTER	F11, J11, G11
PERMIT_GPIO	G20, J20
5V	A25, B25, C25
OPEN_NO	J22, I22
OPEN_GPIO	I27, F27, J27
CLOSED_NO	J24, B24
CLOSED_GPIO	B29, H29, J29
ONEWIRE	I21, F21, G21, H21

Copper check: after jumpers but before resistors/U1, each listed group must read below1ohm (subtract lead resistance); different groups must be open. After components are fitted, resistors and semiconductors intentionally join some nets: do not apply the open-circuit rule to that stage.

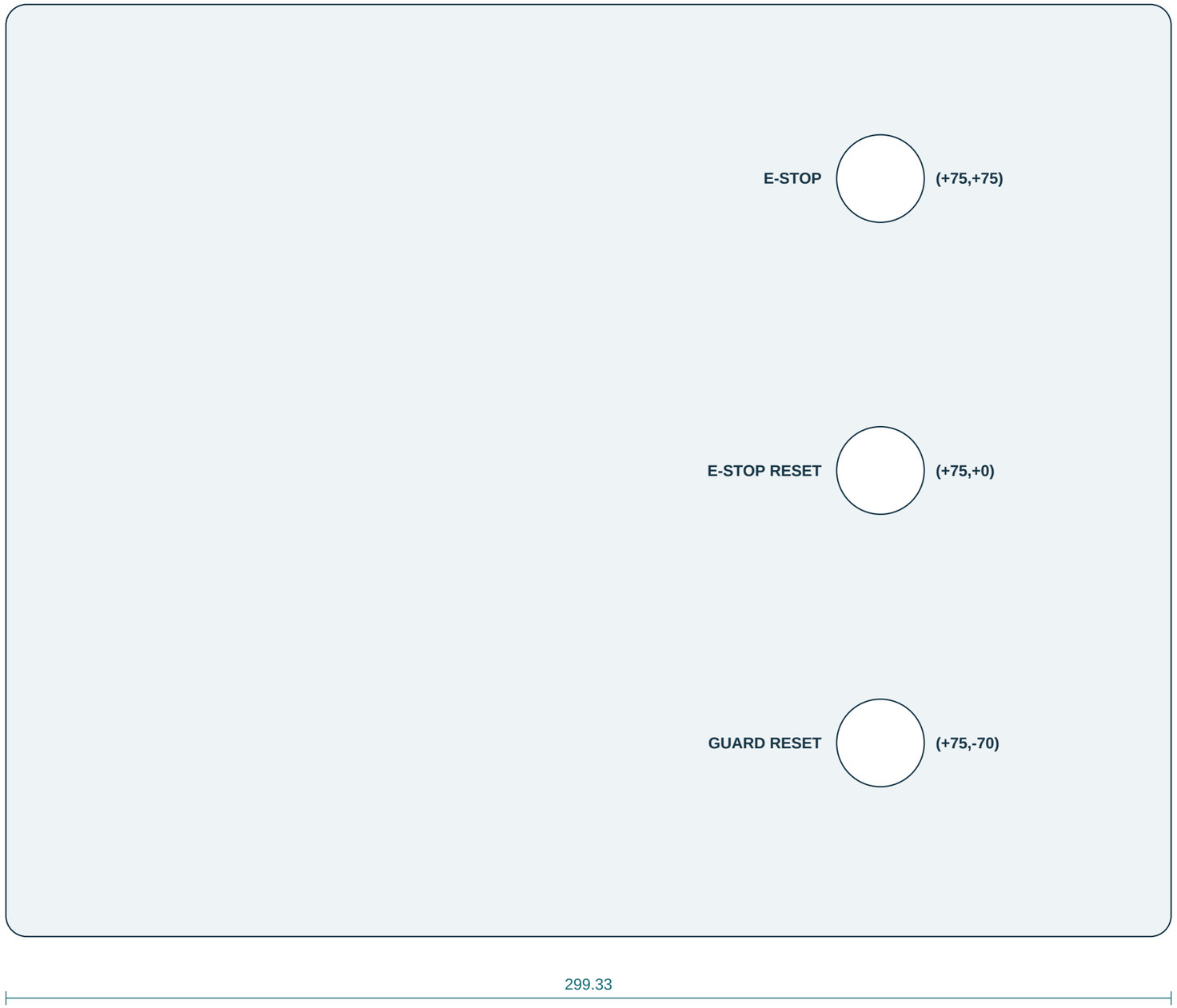
Endpoint test: independently apply5.00..5.25V toA25 and GND toN30 with current limit20mA. Close each actual switch. I27/J29 must each be2.976..3.175V high and below0.2V when open; disconnect either switch wire and confirm low. Verify COM stays at least5.00V with the Pi fully loaded later.

Permit test: supplyP30 with3.3V relativeN30; apply 24V toA5/A11 using a separately isolated current-limited source. J20 must rise above2.0V and remain below3.3V; remove 24V and confirm below0.2V. Verify polarity atU1. R1 is1W: calculated worst dissipation0.326W at26.4V with1% tolerance.

Final connection: remove test supplies before plugging in the Pi. Confirm pin2 has no continuity to any GPIO through a direct wire. Confirm R8 is the only external 1Wire pullup. Record highs/lows at actual Pi header and both probe serials; simulator checks do not satisfy this test.

Cabinet machining / proposed cutout coordinates

Hammond1554YA2GY | datum at outside center | trial assembly and P4 sign-off before cutting



Feature	Location	Cutout
Lid, all3 operators	X=+75;Y=+75,0,-70	D22.5 +0.2/0
Front wall,12V inlet	X=-85;Z=38	D24 +0.1/0
Front wall,24V inlet	X=-35;Z=38	D24 +0.1/0
Each inlet fixing	Offsets(-9.5,+12),(+9.5,-12) from center	2xD3.2
Front wall,guard cable	X=+25;Z=38	D16 +0.2/0;M16

Datums: lid X/Y are measured from its outside bounding-box center, viewed from above. Front-wall X is measured from enclosure center; Z from outside bottom. Front wall is Y=-119.73. Do not use molded lettering as a datum.

Trial fit: install panel and all components on temporary supports. At the inlet centerZ38, verify the actual connector rear body clears the terminal bank, its wiring and the base ribs. The bodies are not included in the simplified allocation view. Record clearance; revise coordinates before cutting if below10mm.

Lid depth: operators are confined to the right-hand zone. Confirm at least10mm free clearance to PCBs, covers and wiring with the lid fully seated. No operator may overlap the relay depth envelope. Protect the gasket groove.

Fixing: remove all electronics before machining. Clamp a sacrificial backing; deburr0.2 max. Fit operator anti-rotation hardware to its current maker drawing. Connector screws: M3 with washer/nut, trim so2threads protrude; do not bottom against internal parts.

This coordinates the known cutouts. Remaining Pi/sensor harness passages and supports must be added to the as-built drawing after connector selection. This is not a complete sealed-enclosure modification release.

Harness routing / service isolation / fuses

External factory AC/DC supplies | all cabinet wiring low voltage | lengths recorded after routing

Circuit	Conductor / termination	Routing / acceptance
12V motor power	18AWG stranded,105C,300V;red positive/black return	F1=0287002.L2A; FHAC holder;keep positive/return paired; do not extend thin TTL leads
24V supply and relay wiring	0.5mm2 stranded,80C min;numbered conductors	F2=0287001.L1A; separate from Pi low-voltage bundle
Guard channels	LAPP1119754,4x0.5mm2;stock1.8m; nominalOD5.7	Cores1/2=11/12;3/4=21/22. LAPP53015130 PG13.5 plastic gland at switch
Guard cable entry	Hammond1427BCGM16, M16x1.5,4..8mm clamp	Cabinet only; never install metal M16 gland in PG13.5 plastic switch
Logic / endpoints / probes	24AWG stranded,60V/80C; E-007 wire destinations	Separate sleeves from power; keep moving lead clear of cam, rails and phone hinge
Screw-clamp ends	One conductor per clamp; matched insulated ferrule	Use calibrated ratchet die; no tinned ends; pull-check every crimp

Prepare: route with lids and guards temporarily fitted, allow service loops without touching moving parts, then cut. Record actual length and wire ID in the as-built schedule. Use screwed cable anchors; adhesive-only anchors are not accepted. Clamp jackets, never individual insulated cores, at enclosure entries. Keep the four guard cores in their own jacket; no intermediate bypass connectors.

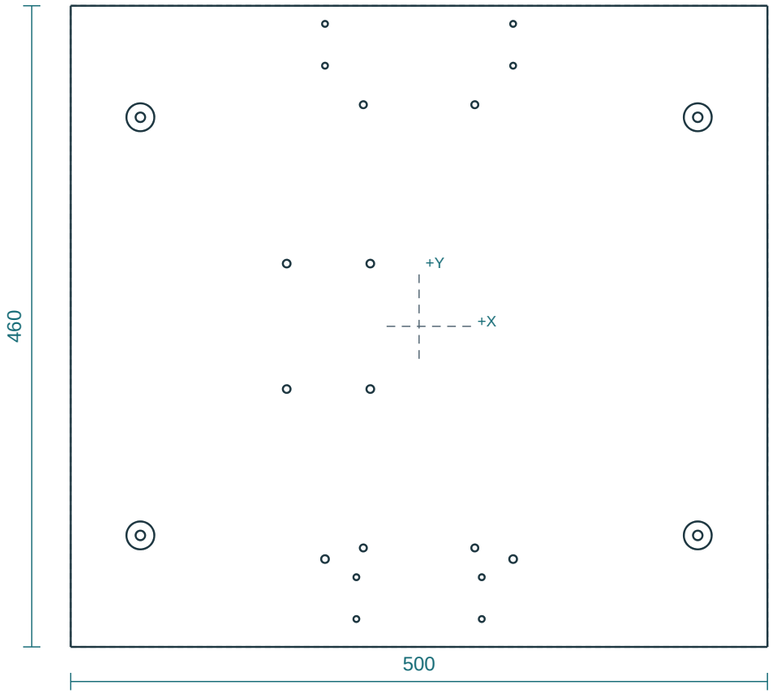
Terminal details: UT2,5 strip9mm,0.5..0.6Nm. D4NS terminals0.6..0.8Nm; use0.5mm2 conductors and the manufacturer's internal lead-length preparation. PG13.5 gland has9mm thread, equal to the switch's9mm maximum. Verify it seats without fouling the internal wiring.

Fuse qualification:2A and1A are prototype selections, not evidence of coordination. Measure normal peak current, inrush and wire temperature. Check the exact supply overload behavior against fuse curves. A current-limited24V supply may never clear a1A fuse; all wiring must withstand its maximum sustained fault output. Do not increase fuse values to cure nuisance opening without a revised review.

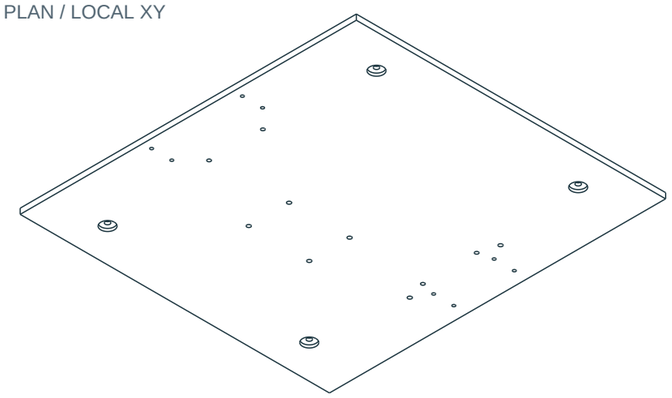
Service isolation: stop motion, support the yoke, then disconnect the AC plugs of the12V,24V and Pi supplies. Keep all three disconnected plugs under the servicing person's exclusive control. Check absence of voltage at X1:1/2,3/4,13/2 and Pi5V/GND before work. If exclusive control is not possible, provide a site-approved lockable isolation method. E-stop is not service isolation. Do not unmate the DC connectors while energized.

Fixture base

Qty 1 | 6061-T6 aluminum | Local part coordinates; see assembly placement



PLAN / LOCAL XY



ISOMETRIC

EDGE / LOCAL XZ

ID	X	Y	Feature
H1	-200	-150	D6.8 THRU; 1/4-20 frame bolt
H2	-200	150	D6.8 THRU; 1/4-20 frame bolt
H3	200	-150	D6.8 THRU; 1/4-20 frame bolt
H4	200	150	D6.8 THRU; 1/4-20 frame bolt
H5	-40	-159	M6 x 1 THRU
H6	-40	159	M6 x 1 THRU
H7	40	-159	M6 x 1 THRU
H8	40	159	M6 x 1 THRU
H9	-95	-45	D5.5 THRU; M5 x 16 from underside
H10	-95	45	D5.5 THRU; M5 x 16 from underside
H11	-35	-45	D5.5 THRU; M5 x 16 from underside
H12	-35	45	D5.5 THRU; M5 x 16 from underside
H13	-45	-210	M5 x 0.8 THRU
H14	-45	-180	M5 x 0.8 THRU
H15	45	-210	M5 x 0.8 THRU
H16	45	-180	M5 x 0.8 THRU
H17	-67.5	-167	D5.5 THRU; M5 x 16 into stop post
H18	67.5	-167	D5.5 THRU; M5 x 16 into stop post
H19	-67.5	187	M5 x 0.8 THRU; endpoint stand
H20	-67.5	217	M5 x 0.8 THRU; endpoint stand
H21	67.5	187	M5 x 0.8 THRU; endpoint stand
H22	67.5	217	M5 x 0.8 THRU; endpoint stand

Blank 500 x 460 x 8.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

Top is assembly datum A (global Z=0).

Mount at global Z=-8. Flatness 0.20 over plate.

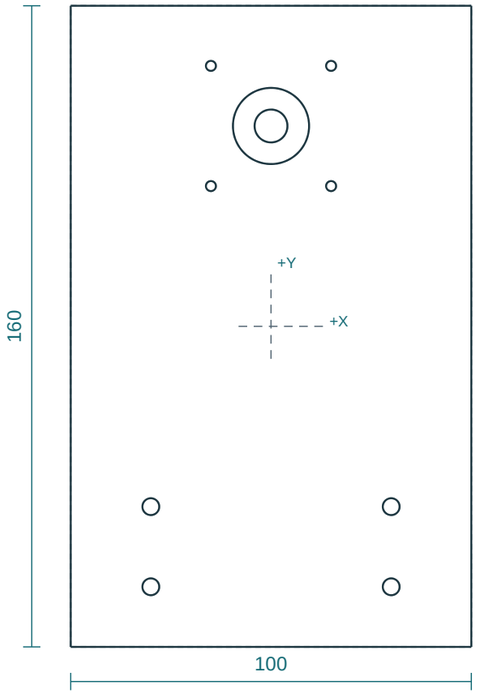
Thread depths are limited to 8 mm; do not bottom bolts.

Four frame mounting holes: counterbore D20 x 4 deep from top for 3393 fasteners and 1 mm washers.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

Bearing tower

Qty 2 | 6061-T6 aluminum | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	0	50	D8.2 THRU; bearing axis
H2	-30	-65	M5 x 0.8 THRU
H3	-30	-45	M5 x 0.8 THRU
H4	30	-65	M5 x 0.8 THRU
H5	30	-45	M5 x 0.8 THRU
H6	-15	35	M3 x 0.5 THRU; retainer
H7	-15	65	M3 x 0.5 THRU; retainer
H8	15	35	M3 x 0.5 THRU; retainer
H9	15	65	M3 x 0.5 THRU; retainer

Blank 100 x 160 x 8.

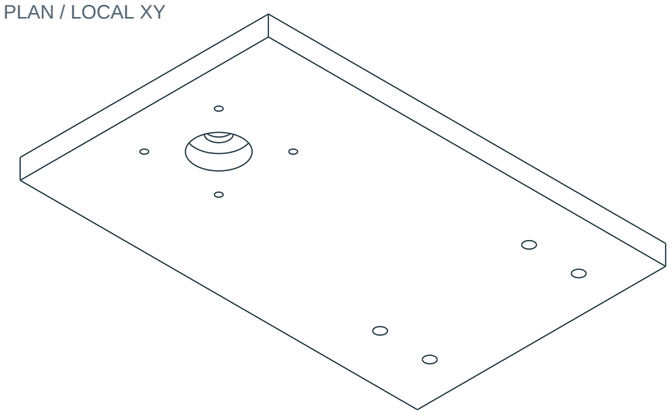
Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

Counterbore D19 H7 (+0.021/0) from Z=8 face, depth 6.05 +0.05/0. Ra 1.6 seat.

Bearing axis true position D0.05 to bottom and side datums; paired tower axes coaxial within D0.05 after assembly.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



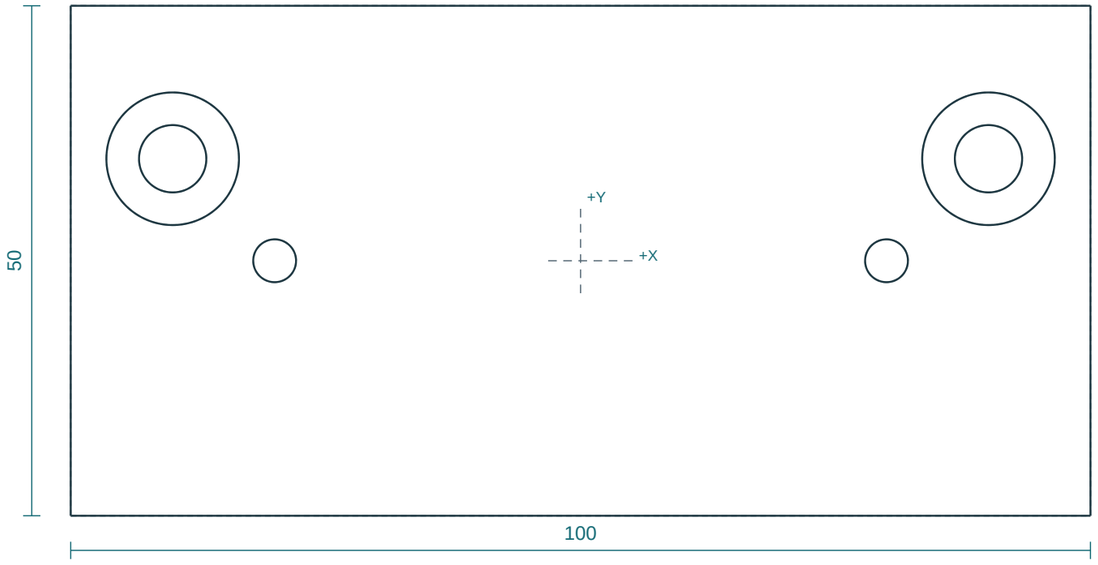
ISOMETRIC



EDGE / LOCAL XZ

Tower sole plate

Qty 2 | 6061-T6 aluminum | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	-40	10	D6.6 THRU; M6 x 16 + washer
H2	40	10	D6.6 THRU; M6 x 16 + washer
H3	-30	0	M5 x 0.8 THRU; angle foot
H4	30	0	M5 x 0.8 THRU; angle foot

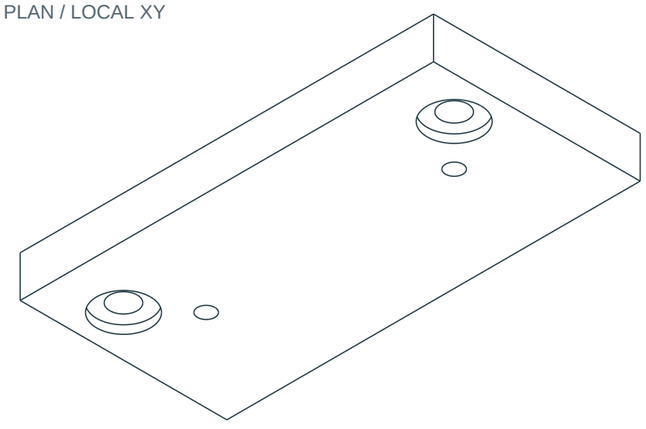
Blank 100 x 50 x 10.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

Two M6 holes: counterbore D13 x 2 deep from top. M6 x 16 + 1 mm washer gives 7 mm engagement into base.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



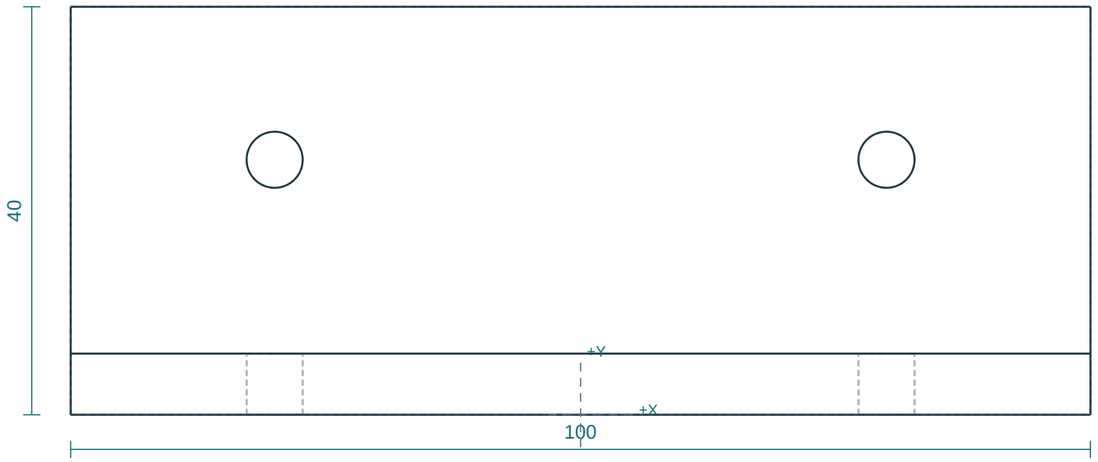
ISOMETRIC



EDGE / LOCAL XZ

Tower angle

Qty 2 | 6061-T6 aluminum | Local part coordinates; see assembly placement



Machine from 100 x 40 x 50 stock; 6 mm legs.

Base holes: X +/-30, Y25, D5.5 THRU along Z.

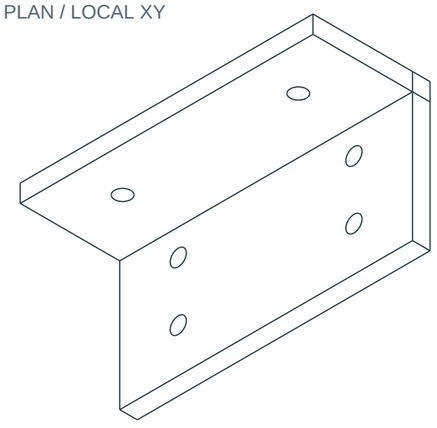
Upright holes: X +/-30, Z15 and 35, D5.5 along Y.

Origin at upright outside / base underside intersection.

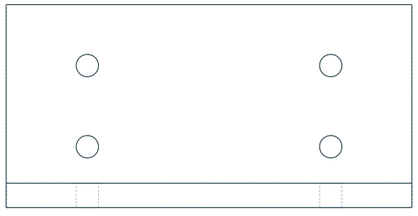
Inside relief R2 max; outside angle 90 +/-0.05 deg.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



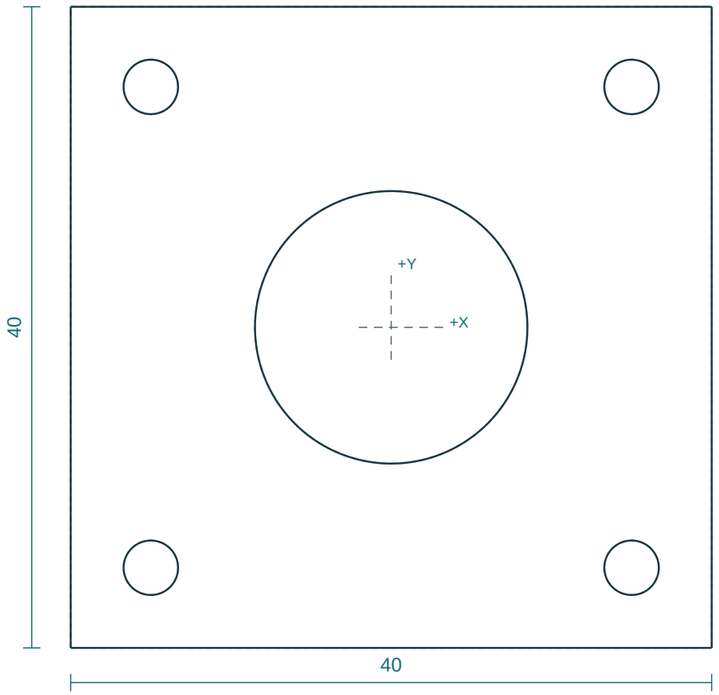
ISOMETRIC



EDGE / LOCAL XZ

Bearing retaining plate

Qty 2 | 6061-T6 aluminum | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	0	0	D17 THRU; retain OUTER race only
H2	-15	-15	D3.4 THRU; M3 x 8
H3	-15	15	D3.4 THRU; M3 x 8
H4	15	-15	D3.4 THRU; M3 x 8
H5	15	15	D3.4 THRU; M3 x 8

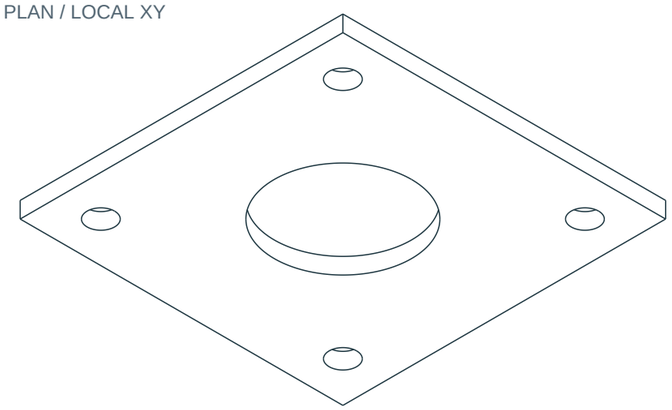
Blank 40 x 40 x 2.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

Do not clamp inner race; check 0.05-0.10 axial outer-race clearance before tightening.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



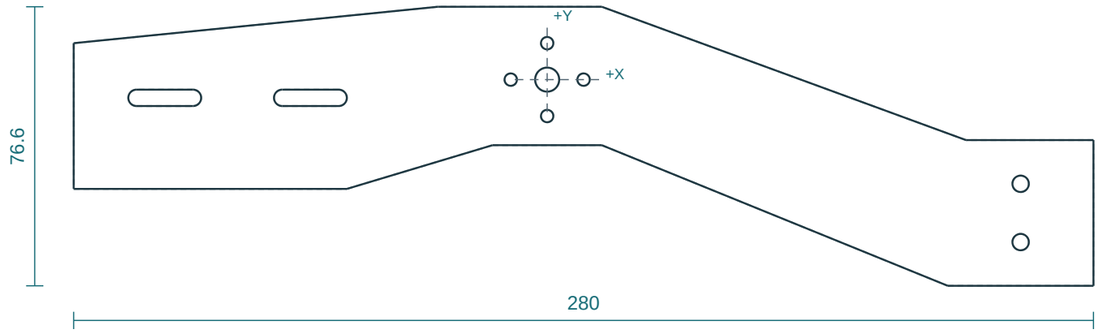
EDGE / LOCAL XZ



ISOMETRIC

Yoke / counterbalance arm

Qty 2 | 6061-T6 aluminum | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	0	0	D6.6 THRU
H2	0	10	D3.4 THRU
H3	0	-10	D3.4 THRU
H4	10	0	D3.4 THRU
H5	-10	0	D3.4 THRU
H6	130	-44.6	D4.5 THRU
H7	130	-28.6	D4.5 THRU
S1	-105	-5	20 x 4.5 slot along X
S2	-65	-5	20 x 4.5 slot along X

6 mm plate. Datum XY=shaft axis; Z=0 outer face.

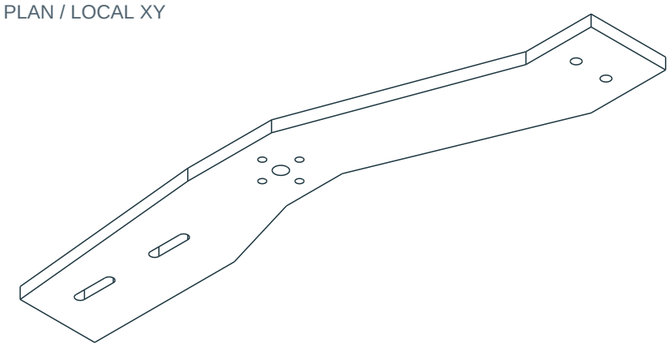
Profile vertices in profile-vertices.json / native STEP.

D0.05 position for the four hub holes; other holes +/-0.10.

Front and rear arm are identical blanks; install mirrored.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY

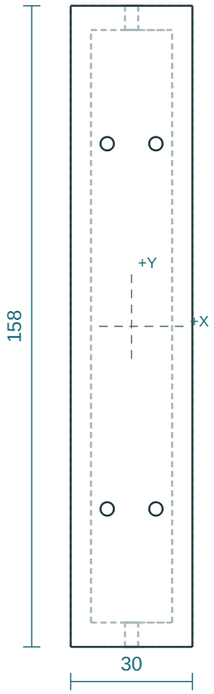


ISOMETRIC

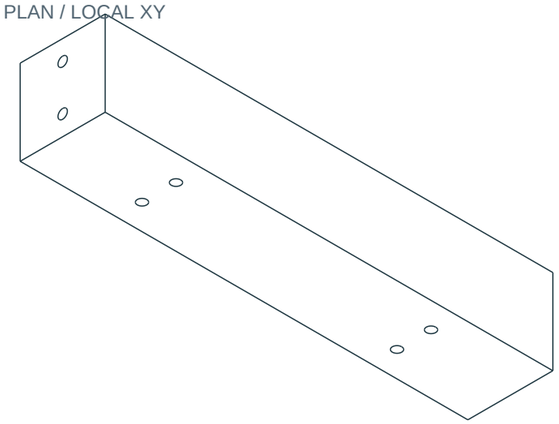
EDGE / LOCAL XZ

Lightened crossbar

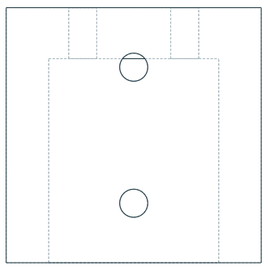
Qty 1 | 6061-T6 aluminum | Local part coordinates; see assembly placement



PLAN / LOCAL XY



ISOMETRIC



EDGE / LOCAL XZ

Blank 30 X x 158 Y x 30 Z; origin at center.

Pocket 20 X x 146 Y x 24 deep from bottom; R3 corners permitted. Model has sharp pocket corners.

Top: 4 x M4 x 0.7 THRU at X +/-6, Y +/-45.

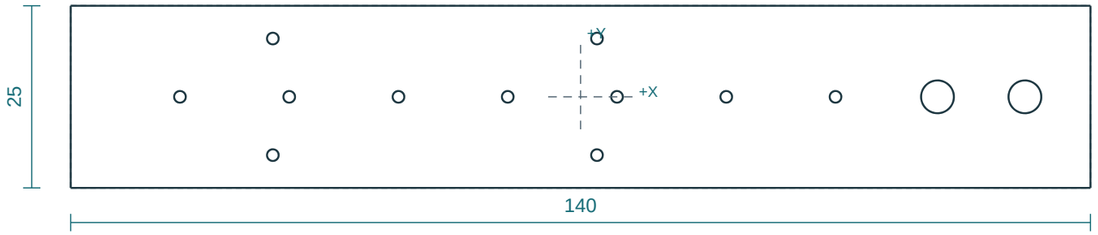
Each end: 2 x M4 x 0.7 THRU at X0, Z +/-8.

Use M4 x 10 through 6 mm arms/strips: 4 mm engagement.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

Guide carrier strip

Qty 2 | 6061-T6 aluminum | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	-55	0	M2 x 0.4 THRU; rail
H2	-40	0	M2 x 0.4 THRU; rail
H3	-25	0	M2 x 0.4 THRU; rail
H4	-10	0	M2 x 0.4 THRU; rail
H5	5	0	M2 x 0.4 THRU; rail
H6	20	0	M2 x 0.4 THRU; rail
H7	35	0	M2 x 0.4 THRU; rail
H8	-42.25	-8	M2 x 0.4 THRU; stop
H9	-42.25	8	M2 x 0.4 THRU; stop
H10	2.25	-8	M2 x 0.4 THRU; stop
H11	2.25	8	M2 x 0.4 THRU; stop
H12	49	0	D4.5 THRU; M4 x 10
H13	61	0	D4.5 THRU; M4 x 10

Blank 140 x 25 x 6.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

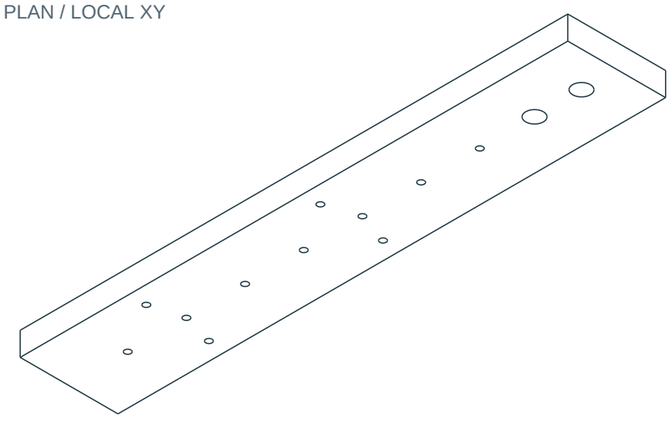
Install global X=75, Y=+/-45, Z=118.4.

Rail holes become X20..110; rail start X15.

Guide stops were re-centered to actual block travel.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



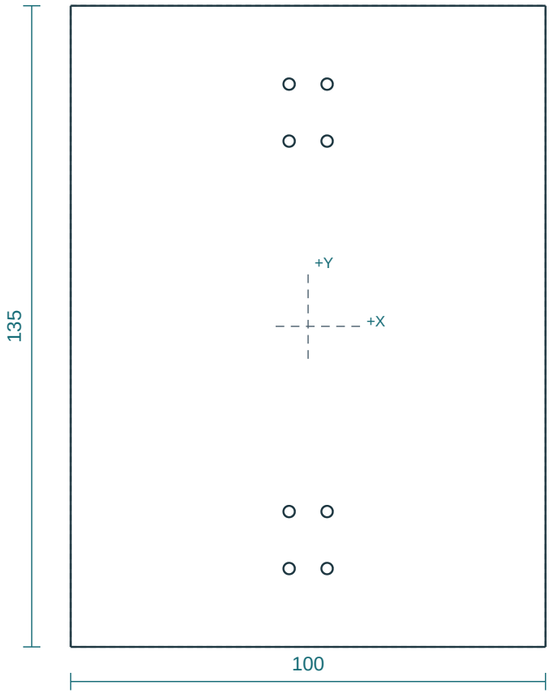
ISOMETRIC



EDGE / LOCAL XZ

Floating contact plate blank

Qty 1 | 6061-T6 aluminum | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	-4	-51	D2.4 THRU; M2 x 6 into MGN7C
H2	-4	-39	D2.4 THRU; M2 x 6 into MGN7C
H3	-4	39	D2.4 THRU; M2 x 6 into MGN7C
H4	-4	51	D2.4 THRU; M2 x 6 into MGN7C
H5	4	-51	D2.4 THRU; M2 x 6 into MGN7C
H6	4	-39	D2.4 THRU; M2 x 6 into MGN7C
H7	4	39	D2.4 THRU; M2 x 6 into MGN7C
H8	4	51	D2.4 THRU; M2 x 6 into MGN7C

Blank 100 x 135 x 4.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

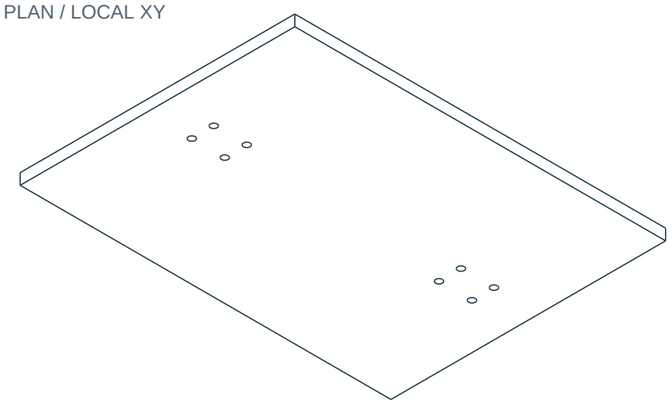
Install center global X55, Y0, bottom Z132.4.

4 mm plate + M2 x 6 gives 2 mm block engagement.

Phone-contact inserts and restraint holes are HOLD P1 until actual phone is measured.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



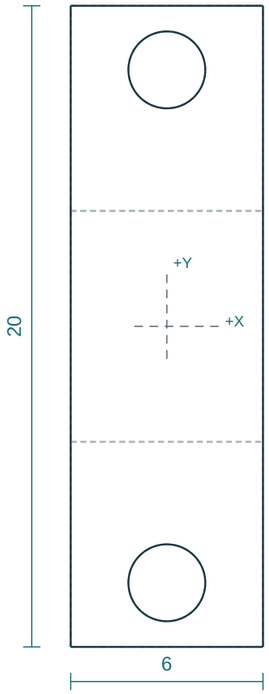
ISOMETRIC



EDGE / LOCAL XZ

Guide travel stop

Qty 4 | POM-C acetal | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	0	-8	D2.4 THRU; M2 x 12
H2	0	8	D2.4 THRU; M2 x 12

Blank 6 x 20 x 8.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

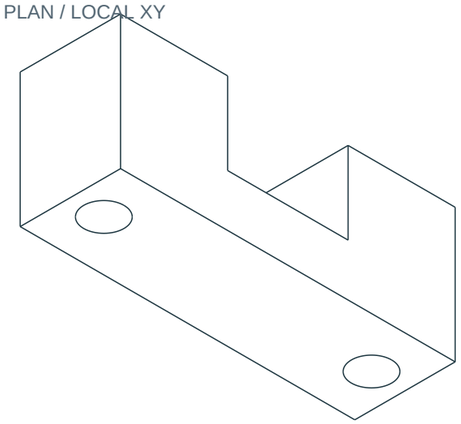
Underside channel 6 long x 7.2 wide x 4.9 deep, centered Y=0, clears rail.

Bond 2 mm silicone pad to inward X face.

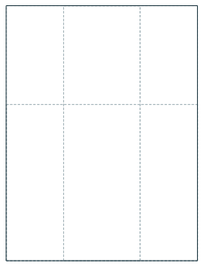
Stop body centers X=32.75 and 77.25 on each rail; nominal float +/-6 about X55.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



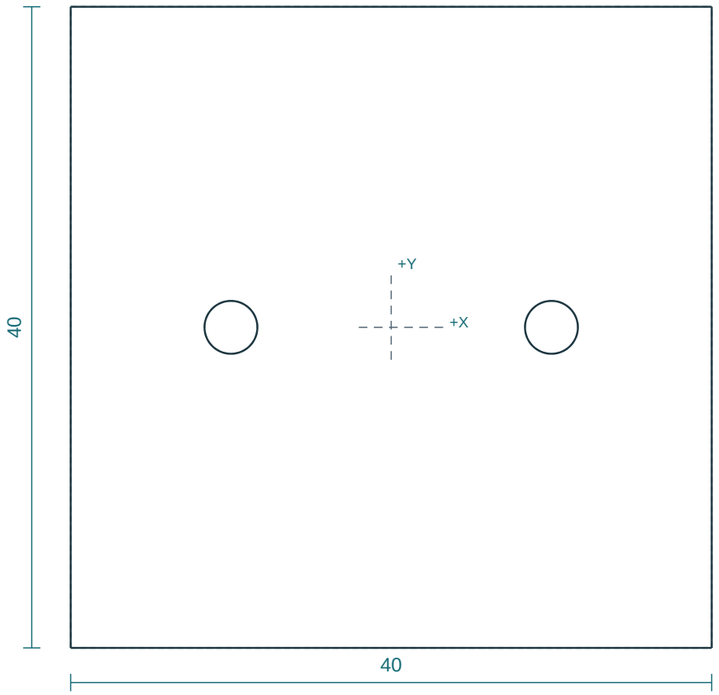
ISOMETRIC



EDGE / LOCAL XZ

Counterweight

Qty 2 | Low-carbon steel; zinc plate after machining | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	-10	0	M4 x 0.7 THRU
H2	10	0	M4 x 0.7 THRU

Blank 40 x 40 x 25.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

Nominal 0.31 kg each; weigh finished part.

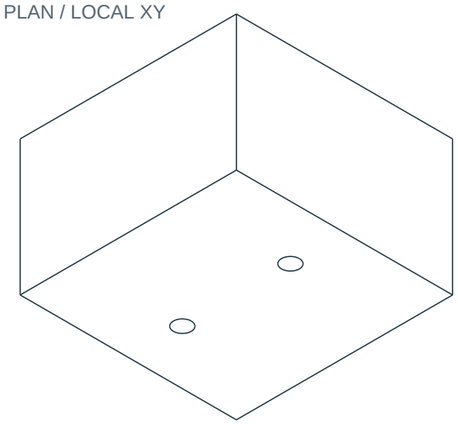
Initial center X=-85, Z=165; adjust on B120.

Use M4 x 16 + 1 mm washer through 4 mm B120: 11 mm thread engagement.

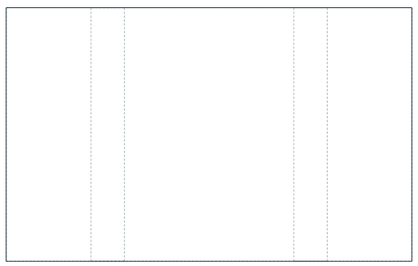
Balance both X and Z moments; verify Q04 sweep.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



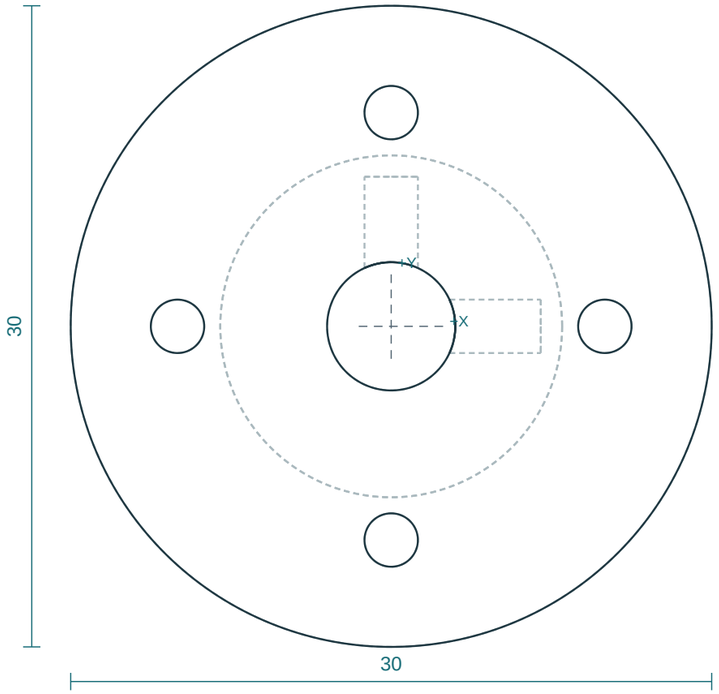
ISOMETRIC



EDGE / LOCAL XZ

Output flange hub

Qty 2 | 6061-T6 aluminum | Local part coordinates; see assembly placement



Origin: barrel end center; shaft axis Z.

Barrel D16 x 10; flange D30 x 6; bore D6 H7.

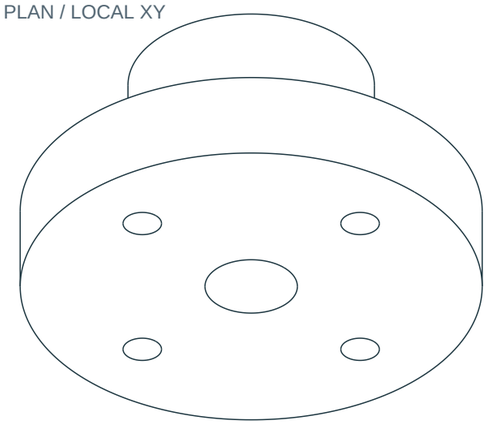
Flange 4 x M3 x 0.5 THRU on PCD20, at 0/90/180/270 deg.

Two radial M3 x 0.5 THRU at Z5; axes +X, +Y.

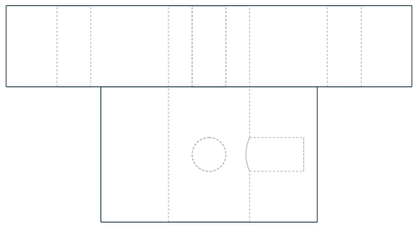
M3 x 4 flat-point screws contact shaft flats; proof torque 0.75 N m off-phone. Do not use adhesive as retention.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



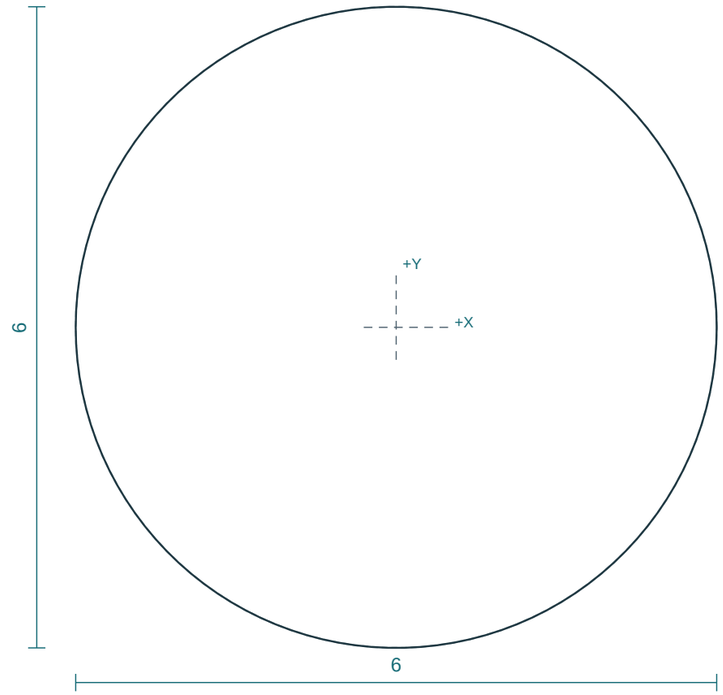
ISOMETRIC



EDGE / LOCAL XZ

Ground output shaft

Qty 1 | Ground stainless steel, D6 g6 | Local part coordinates; see assembly placement



PLAN / LOCAL XY



ISOMETRIC



EDGE / LOCAL XZ

Cut length 108 +/-0.2; chamfer 0.2 x 45 deg.

Shaft D6 g6: 5.988..5.996 mm; no plating on fits.

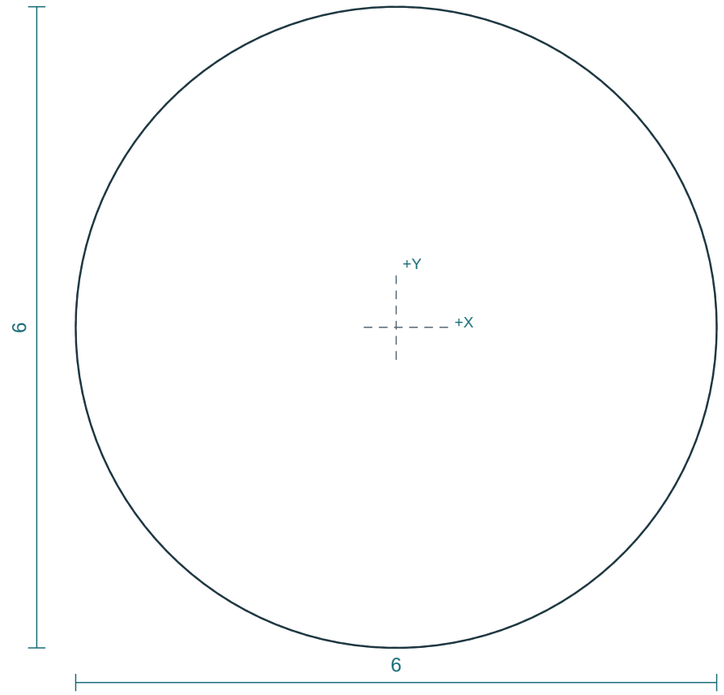
Machine two perpendicular flats, 0.30 deep, 8 long, at hub set-screw station shown in axial stack.

Model omits clocked flats until clutch datum is confirmed (HOLD P2).

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

Ground output shaft

Qty 1 | Ground stainless steel, D6 g6 | Local part coordinates; see assembly placement



PLAN / LOCAL XY



ISOMETRIC



EDGE / LOCAL XZ

Cut length 84 +/-0.2; chamfer 0.2 x 45 deg.

Shaft D6 g6: 5.988..5.996 mm; no plating on fits.

Machine two perpendicular flats, 0.30 deep, 8 long, at hub set-screw station shown in axial stack.

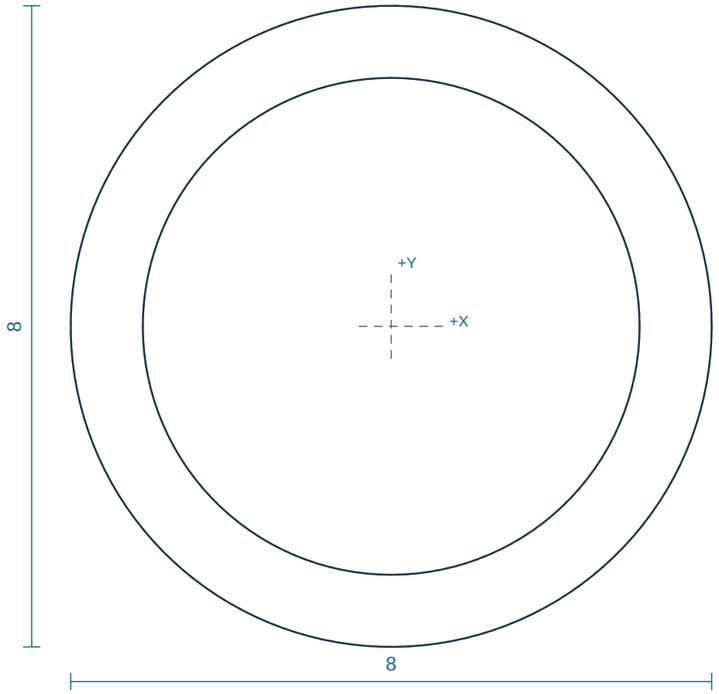
Model omits clocked flats until clutch datum is confirmed (HOLD P2).

Additional sensing-cam flats: two perpendicular0.30-deep x8-long flats centered at localZ71 (globalY156). Clock to the assembled cam; do not cut into a bearing seat.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

Inner-race spacer

Qty 2 | 303 stainless | Local part coordinates; see assembly placement

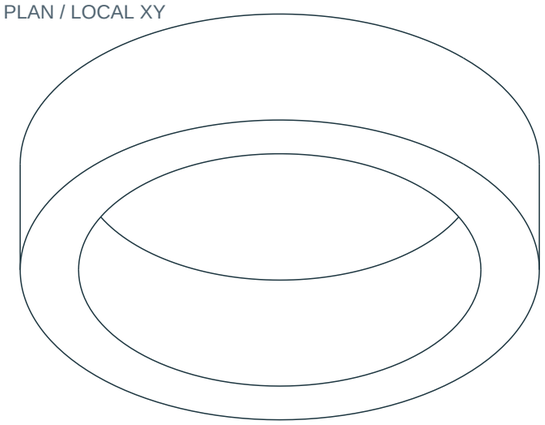


D8 OD x D6.2 ID x 2 +/-0.02 long.

Faces parallel 0.02; no burrs. Contact inner race only.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



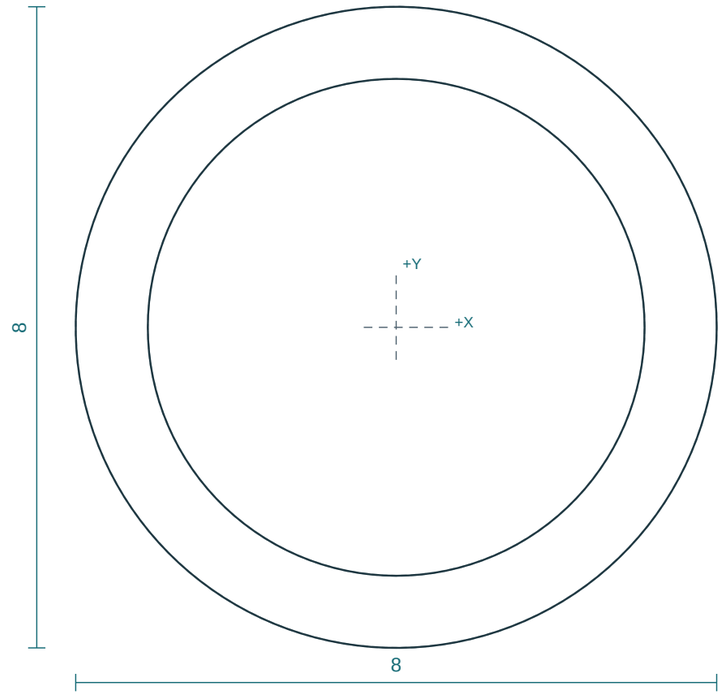
ISOMETRIC



EDGE / LOCAL XZ

Inner-race spacer

Qty 2 | 303 stainless | Local part coordinates; see assembly placement

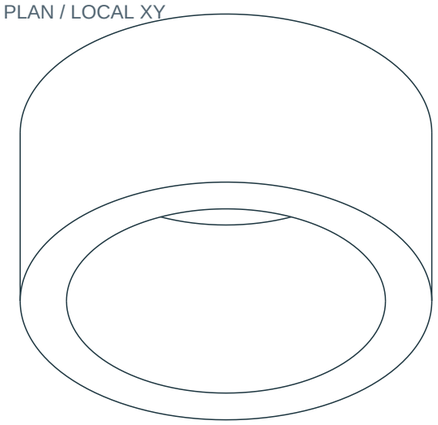


D8 OD x D6.2 ID x 4 +/-0.02 long.

Faces parallel 0.02; no burrs. Contact inner race only.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



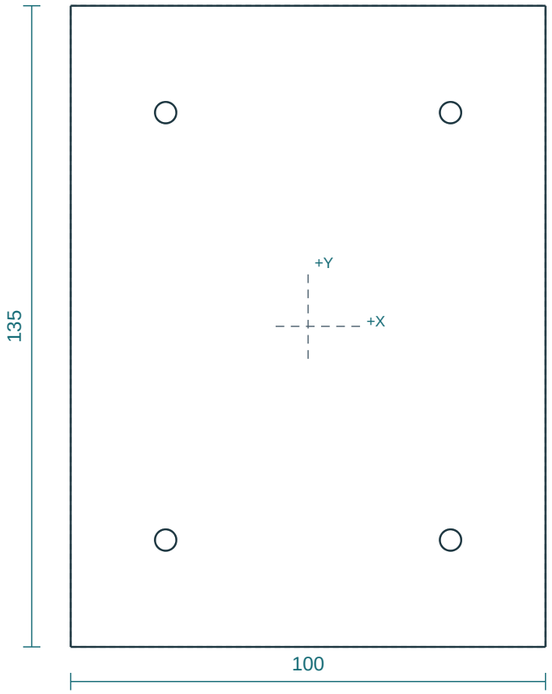
ISOMETRIC



EDGE / LOCAL XZ

Fixed tray

Qty 1 | 6061-T6 aluminum | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	-30	-45	D4.5 THRU; M4 x 12 into posts
H2	-30	45	D4.5 THRU; M4 x 12 into posts
H3	30	-45	D4.5 THRU; M4 x 12 into posts
H4	30	45	D4.5 THRU; M4 x 12 into posts

Blank 100 x 135 x 6.

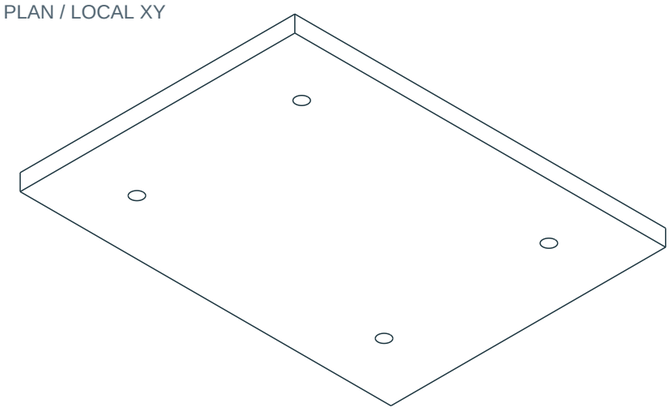
Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

Install X=-65, Y0, Z130.4. 1 mm pad on top.

Actual phone restraint/insert mounting: HOLD P1.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



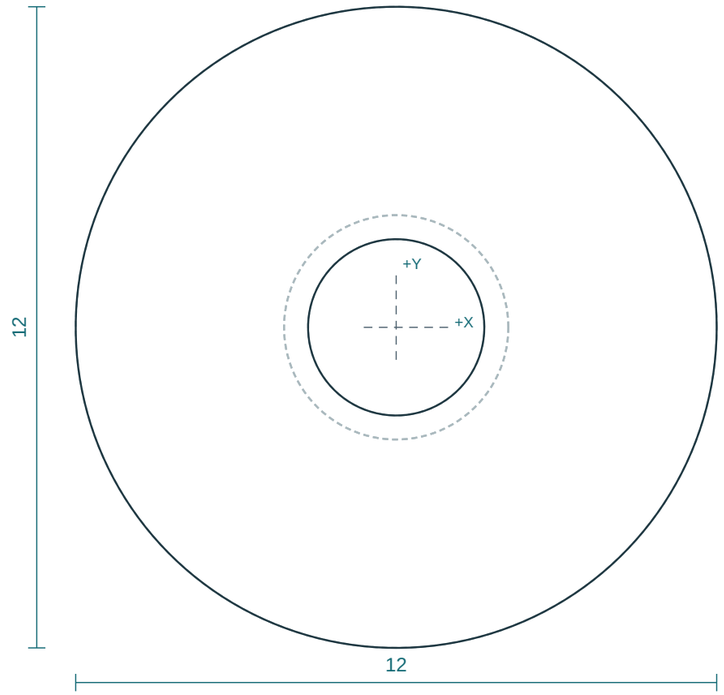
EDGE / LOCAL XZ



ISOMETRIC

Fixed tray standoff

Qty 4 | 6061-T6 aluminum | Local part coordinates; see assembly placement



PLAN / LOCAL XY



ISOMETRIC



EDGE / LOCAL XZ

D12 x 130.4 +/-0.05; top M4 x 0.7, 10 deep.

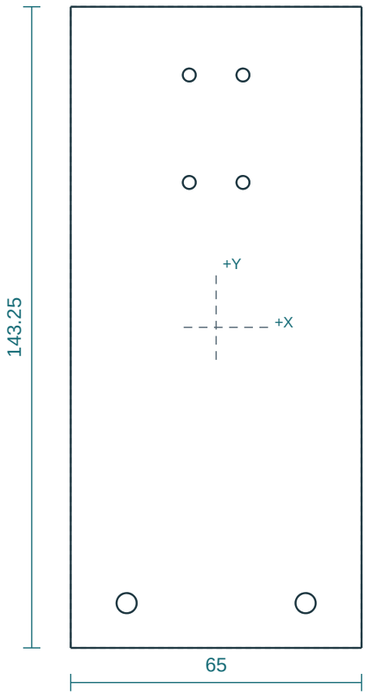
Bottom M5 x 0.8, 10 deep; bottom fasteners from below base require base holes D5.5, not M5 tapped.

Corrected base-hole callout is applied automatically.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

Motor mounting side cheek

Qty 2 | 6061-T6 aluminum | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	-6	56.375	D2.9 THRU; M2.5 x 8, 2 mm engagement
H2	-6	32.375	D2.9 THRU; M2.5 x 8, 2 mm engagement
H3	6	56.375	D2.9 THRU; M2.5 x 8, 2 mm engagement
H4	6	32.375	D2.9 THRU; M2.5 x 8, 2 mm engagement
H5	-20	-61.625	D4.5 THRU; M4 x 16 into B121
H6	20	-61.625	D4.5 THRU; M4 x 16 into B121

Blank 65 x 143.25 x 6.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

Side holes follow ROBOTIS drawing: 12 x 24 pattern; top row 4 below output axis.

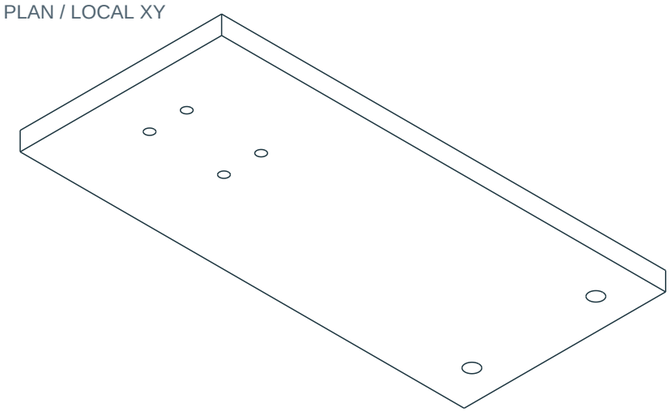
Cheek bottom global Z=8; output axis global Z=140.

Motor depth datum is 17 behind horn face; side holes are 11 and 23 behind front body face.

Two cheeks fit motor 28.5 mm width; shim at body if needed without exceeding 3 mm screw engagement.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



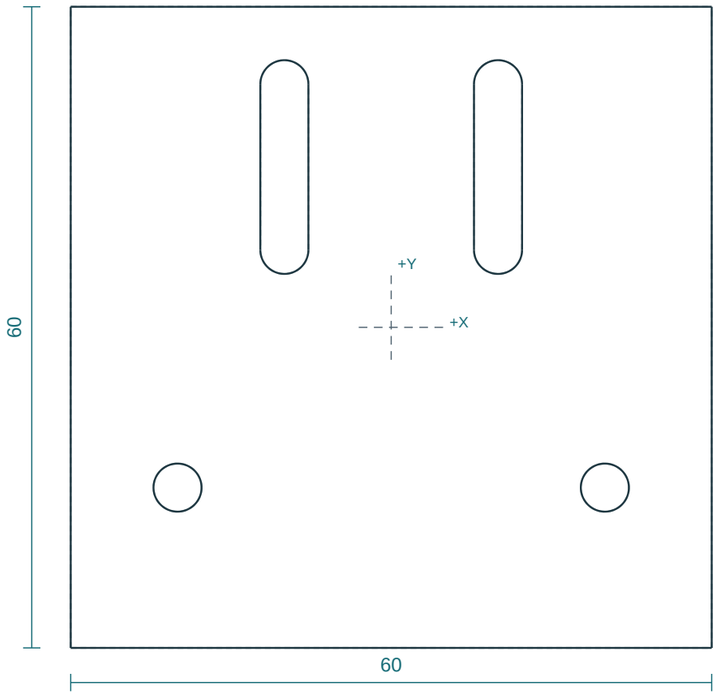
ISOMETRIC



EDGE / LOCAL XZ

Counterweight adjustment plate

Qty 2 | 6061-T6 aluminum | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	-20	-15	D4.5 THRU; M4 x 20 + washer + locknut
H2	20	-15	D4.5 THRU; M4 x 20 + washer + locknut
S1	-10	15	20 x 4.5 vertical slot; M4 x 16
S2	10	15	20 x 4.5 vertical slot; M4 x 16

Blank 60 x 60 x 4.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

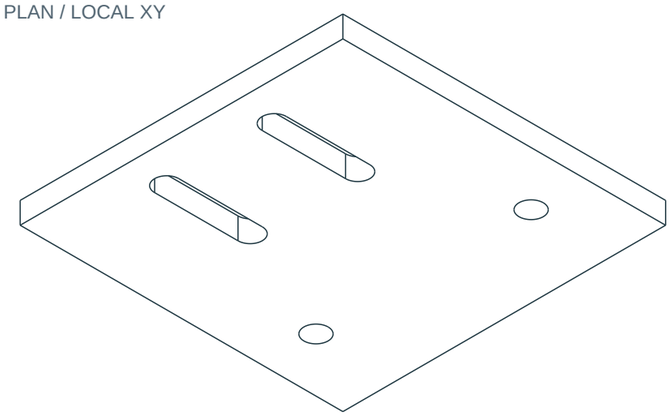
Install center X=-85, Z=150, on arm outer face.

Arm slots provide X travel +/-7.75; upper slots provide Z travel +/-7.75. Adjust both weights equally.

Do not enlarge slots if actual balance lies outside range; revise weight mass and recheck swept clearance.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



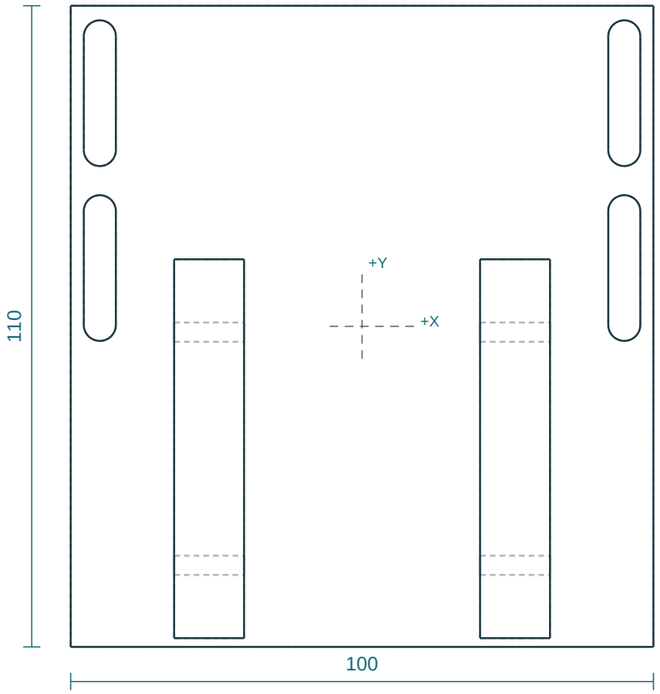
ISOMETRIC



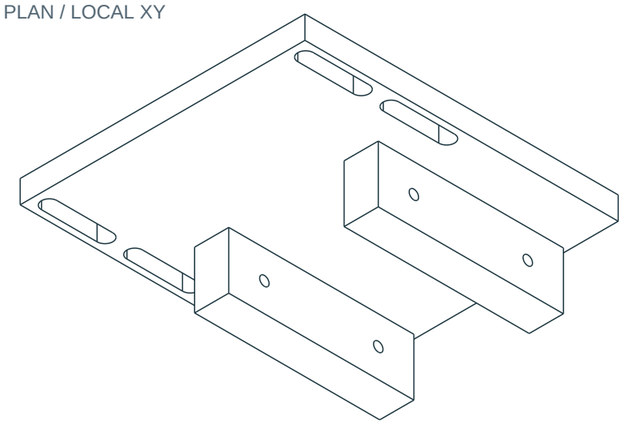
EDGE / LOCAL XZ

Motor adjustable pedestal

Qty 1 | 6061-T6 aluminum | Local part coordinates; see assembly placement



PLAN / LOCAL XY



ISOMETRIC



EDGE / LOCAL XZ

Origin XY center of 100 x 110 sole; sole Z0..8.

Two integral lugs 12 X x 65 Y x 20 high above sole; centers X+/-26.25, Y=-21, global local Z18.

Lug bores 4 x M4 THRU along X at Y=-41/-1, Z18.

4 slots 25 x 5.5 along Y, centers X+/-45, Y10/40.

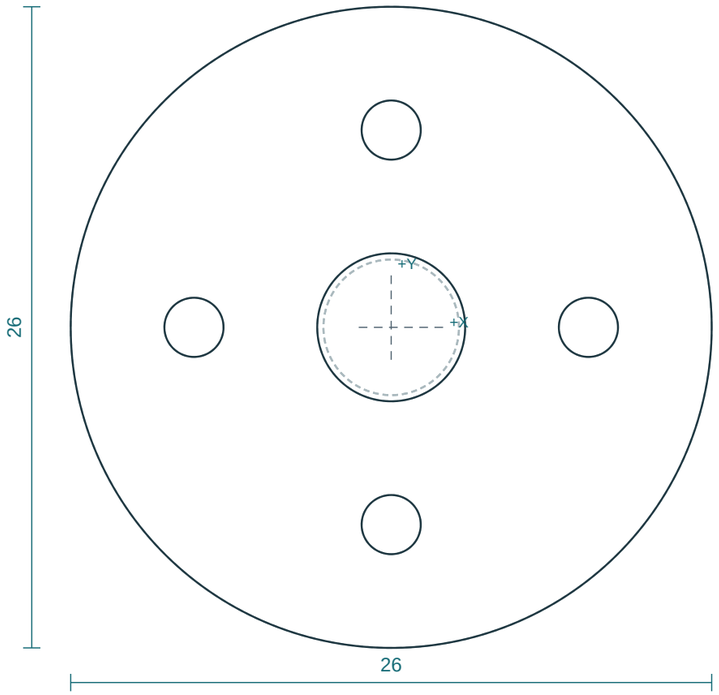
Install global Y=-220; M5 x 16 + 1 mm washer through 8 mm sole gives 7 mm engagement in 8 mm base.

Usable axial adjustment -3 to +9.75 (guard limits forward travel); confirm clutch insertion before tightening. Root relief R2 allowed.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

Integral motor input flange / shaft

Qty 1 | 303 stainless steel | Local part coordinates; see assembly placement



D26 flange x 6; integral D6 g6 shaft x 12 above flange.

4 x D2.4 THRU, PCD16 at 0/90/180/270 degrees.

Underside central relief D5.5 x 3 deep; check central horn bolt clears relief by >=0.5 before installation.

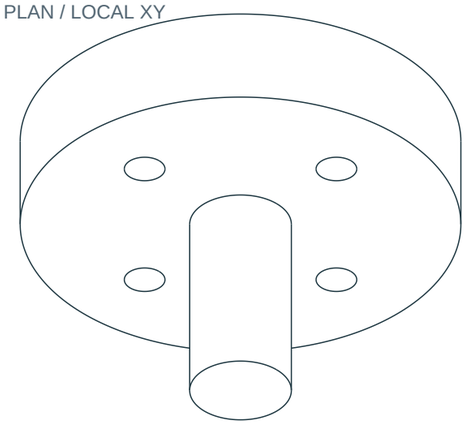
M2 x 8 + 0.2 mm shim washer gives 1.8 mm engagement in horn; maximum allowed by ROBOTIS is 2.0 mm.

Do not loosen the central horn retaining bolt.

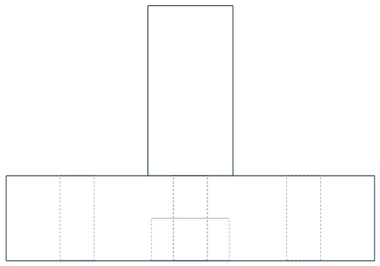
Clutch insertion length and set-screw flat: HOLD P2.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



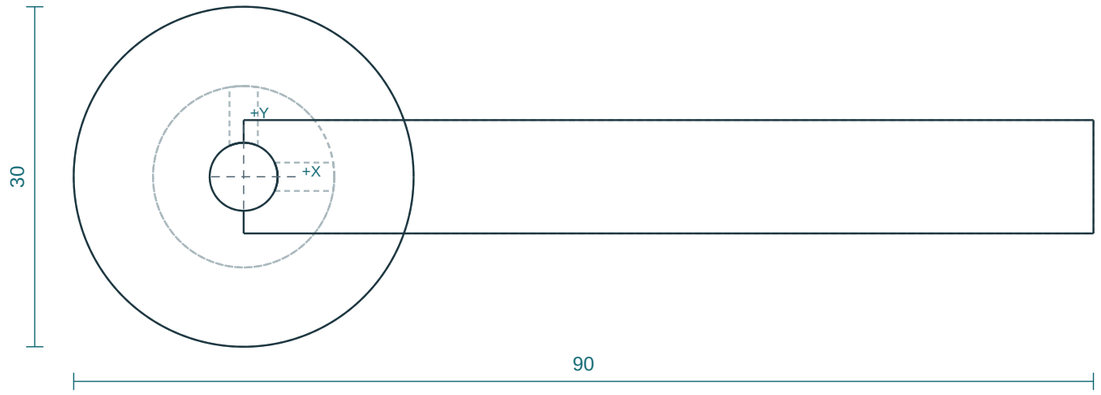
ISOMETRIC



EDGE / LOCAL XZ

Output angular-stop cam

Qty 1 | 6061-T6 aluminum | Local part coordinates; see assembly placement



Barrel D16 x 10; flange D30 x 6 at Z10..16; bore D6 H7.

Integral radial tab X0..75, Y+/-5, Z10..16.

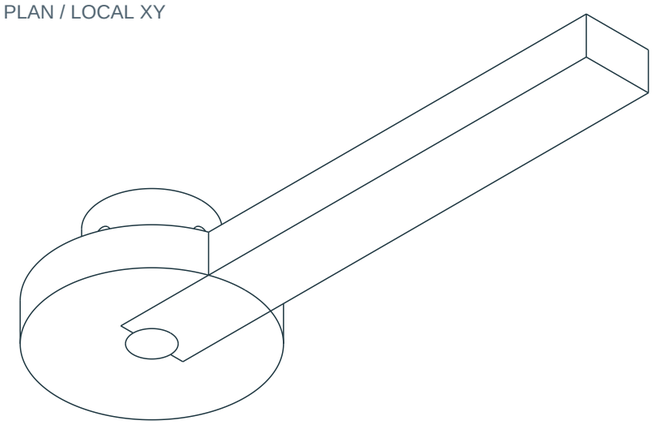
Two radial M3 x 0.5 at Z5, directions +X and +Y.

Two M3 x 4 flat-point screws on shaft flats; proof at 0.75 N m off-phone.

Installed tab points +X at fixture zero degrees. Cam origin global (0,-180,140), axis along +Y.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



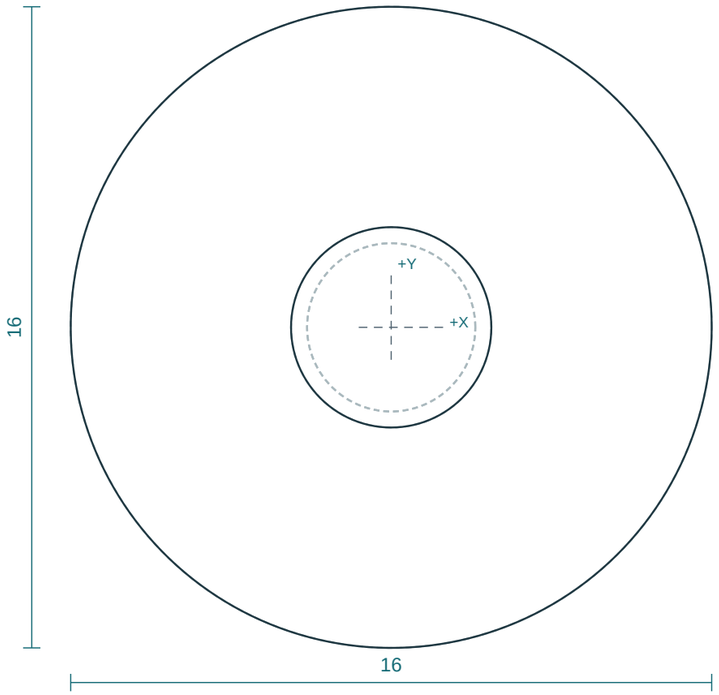
ISOMETRIC



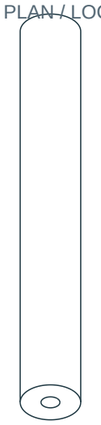
EDGE / LOCAL XZ

Angular-stop post

Qty 2 | 303 stainless steel | Local part coordinates; see assembly placement



PLAN / LOCAL XY



ISOMETRIC



EDGE / LOCAL XZ

D16 x 120; top M6 x 1, 25 deep; bottom M5 x 0.8, 10 deep.

Install X+/-67.5, Y=-167 on base. M5 x 16 from underside.

Top M6 x 25 cap screw + M6 jam nut; 1 mm silicone pad bonded to cap head. Model contact height 135.00 mm for 0/180 degree backup stops; normal motion 5..175.

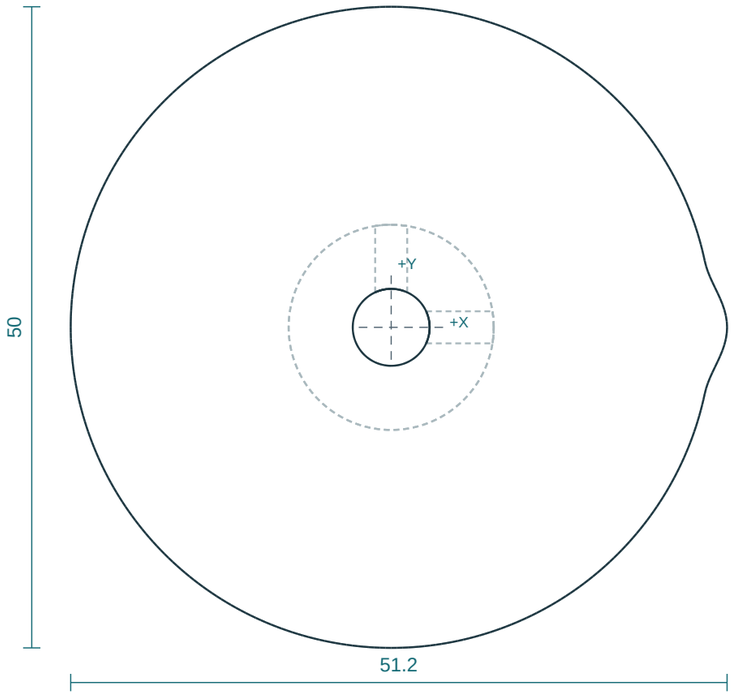
Lock nut after setting; minimum screw engagement 10 mm.

Mechanical stops are overload backups, not cycle targets.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

Endpoint sensing cam

Qty 1 | POM-C acetal | Local part coordinates; see assembly placement



Barrel D16 x10; disc6 thick at Z10..16; bore D6 H7.

Polar profile: R25 outside +/-12deg about +X. Within that sector R=25+0.6*(1+cos(pi*angle/12deg)). Max R26.2.

STEP samples profile each1deg; finish follower surface Ra0.8, remove facets/burrs without changing radius >0.03.

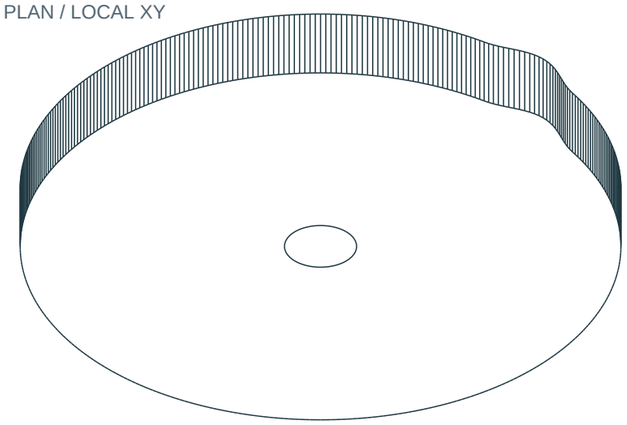
Two radial M3 x0.5 at Z5; M3x4 flat-point screws. Light initial torque0.10Nm; no holding/safety credit.

Origin global(0,151,140), axis+Y. Peak points+X at open. Disc global Y161..167, shaft engagement16mm.

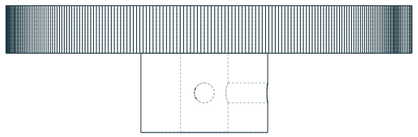
Adjust each switch separately. At peak, measure actual overtravel0.35..0.45mm after electrical make; never exceed the purchased switch's allowed travel. Physical test P3.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



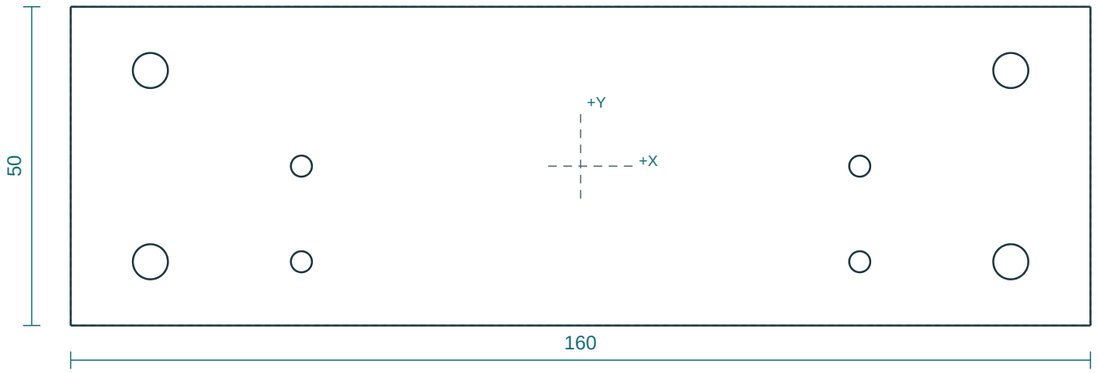
ISOMETRIC



EDGE / LOCAL XZ

Endpoint stand sole

Qty 1 | 6061-T6 aluminum | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	-67.5	-15	D5.5 THRU; M5 x20 +1mm washer
H2	-67.5	15	D5.5 THRU; M5 x20 +1mm washer
H3	67.5	-15	D5.5 THRU; M5 x20 +1mm washer
H4	67.5	15	D5.5 THRU; M5 x20 +1mm washer
H5	-43.8	-15	M4 x0.7 THRU; upright
H6	-43.8	0	M4 x0.7 THRU; upright
H7	43.8	-15	M4 x0.7 THRU; upright
H8	43.8	0	M4 x0.7 THRU; upright

Blank 160 x 50 x 12.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

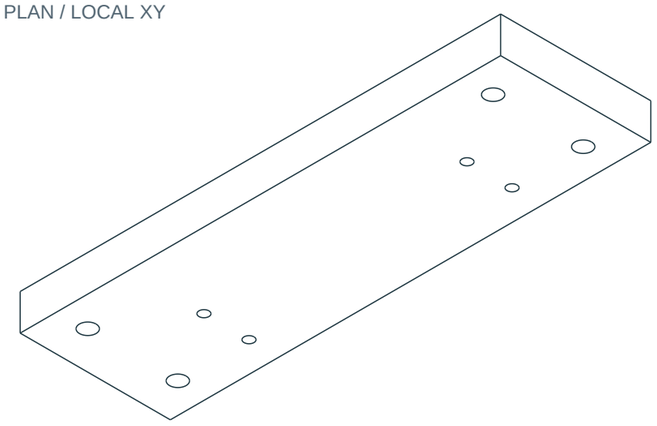
Install center(0,202), bottomZ0. Four M5 base screws.

Upright feet startY167; foot holesY187/202.

Front edgeY177 clears rear tower sole by3mm. Upright foot bottomZ12 clears tower sole topZ10 by2mm.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



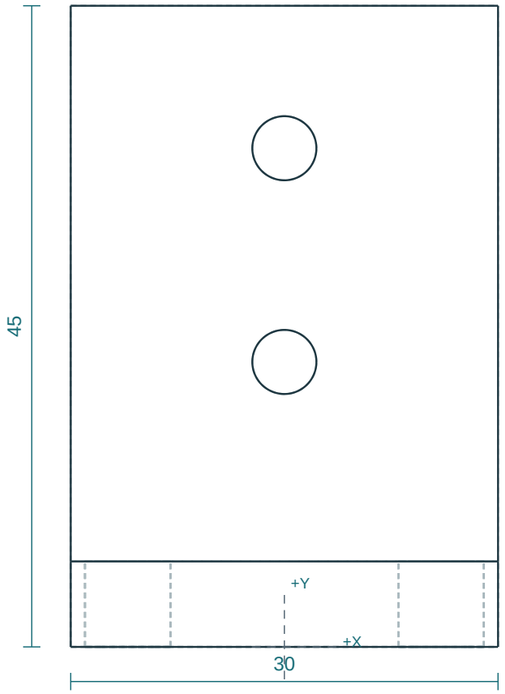
ISOMETRIC



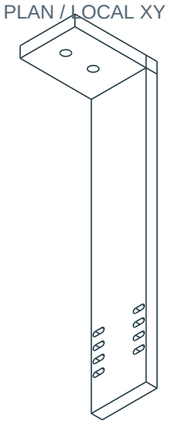
EDGE / LOCAL XZ

Endpoint switch upright

Qty 2 | 6061-T6 aluminum | Local part coordinates; see assembly placement



PLAN / LOCAL XY



ISOMETRIC



EDGE / LOCAL XZ

ID	X	Y	Feature
S1	-11	118.35	6x2.4 slot alongX; Y column=Z
S2	-11	124.85	6x2.4 slot alongX; Y column=Z
S3	-11	131.15	6x2.4 slot alongX; Y column=Z
S4	-11	137.65	6x2.4 slot alongX; Y column=Z
S5	11	118.35	6x2.4 slot alongX; Y column=Z
S6	11	124.85	6x2.4 slot alongX; Y column=Z
S7	11	131.15	6x2.4 slot alongX; Y column=Z
S8	11	137.65	6x2.4 slot alongX; Y column=Z

Machine angle: width30; footY0..45,t6; uprightY0..6,H155.

Local origin: Xmid-width,Yfront,Zfoot bottom. Slot chart uses X,Z coordinates, not top-view X,Y.

Foot holes D4.5 THRU at X0,Y20/35. Use M4x12 with1mm washers:5mm engagement in B126.

Install X+/-43.8,Y167,Z12. FaceY167 supports switches.

Right switch: inward lower pair, global X32.8, Z130.35/136.85. Left: inward upper pair,X-32.8, Z143.15/149.65. Other slots unused.

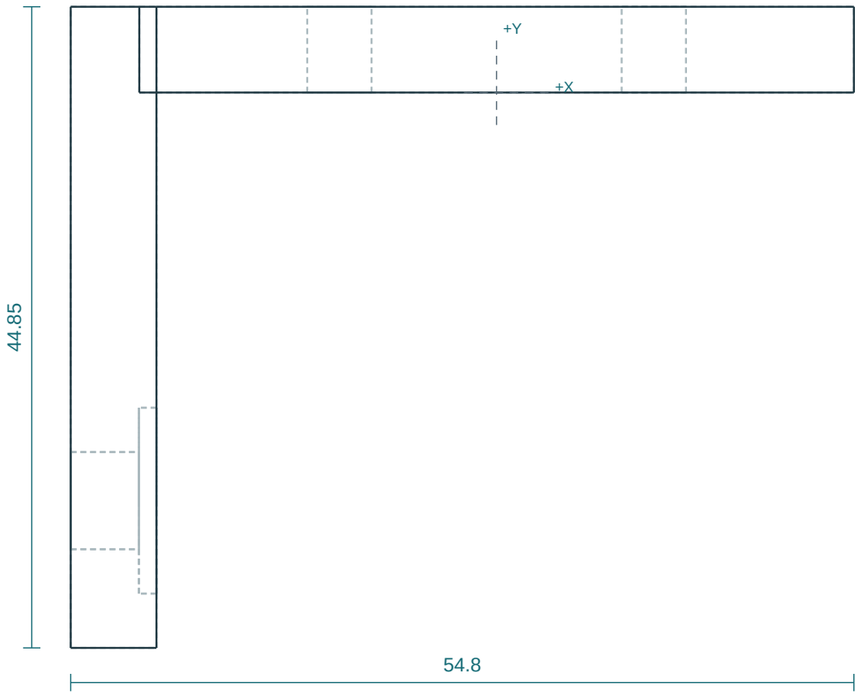
M2x16 through switch and upright; plain washer at head, spring washer and hex nut behind upright; torque0.08..0.10Nm.

Slots give+/-1.8mm radial setup. Lock after continuity and overtravel measurements; never clamp across lever.

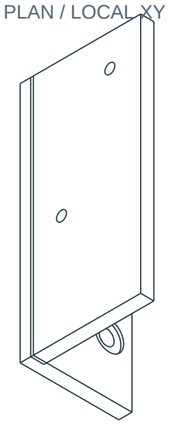
General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

Guard-switch post bracket

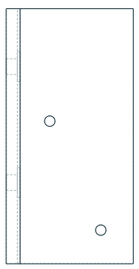
Qty 1 | 6061-T6 aluminum | Local part coordinates; see assembly placement



PLAN / LOCAL XY



ISOMETRIC



EDGE / LOCAL XZ

ID	X	Y	Feature
H1	-11	61.5	D4.5 throughY; chart Y=localZ
H2	11	14.5	D4.5 throughY; chart Y=localZ

Local X=plate midwidth,Y=switch seating face,Z=bottom. Back plate X+/-25,Y0..6,Z0..110.

Side leg X-29.8..-23.8,Y-38.85..6,Z0..110. Inside root radius<=3 allowed; keep switch seating face flat.

Side holes alongX: D6.8 at Y-28.55,Z35/85; counterbore D13x1.225 fromX=-23.8 face. Effective clamp thickness4.775 matches4119/3393 stack.

Install origin(-275,-238.15,255). Side leg seats on left-front post inner faceX=-304.8. Two3393 sets.

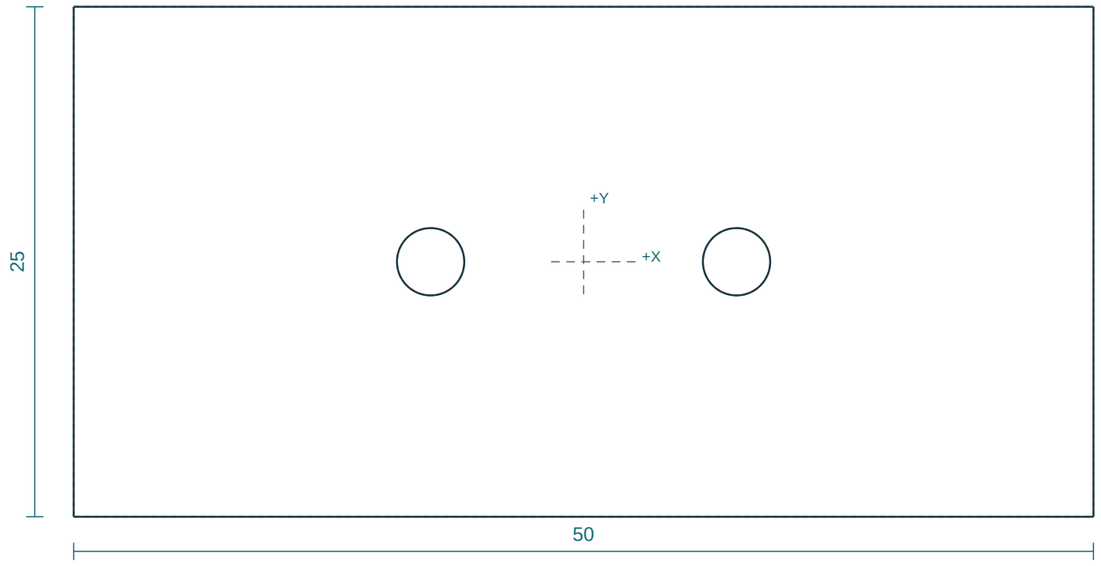
Switch diagonal M4 pitch22x47; body torque0.5..0.7Nm. Do not use the optional locating-stud holes as bolt holes.

Cut two M4x50 blanks to measured device/bracket/washer/nut stack +two full threads; deburr and record final length.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

Guard-key backing plate

Qty 1 | 6061-T6 aluminum | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	-7.5	0	M4 x0.7 THRU; key15mm pitch
H2	7.5	0	M4 x0.7 THRU; key15mm pitch

Blank 50 x 25 x 6.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

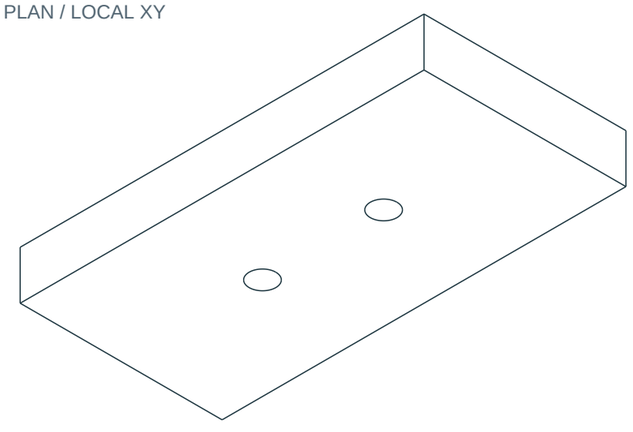
Outside front panel; centerX=-275,Z344; inner faceY=-283.4, outer faceY=-289.4.

Two M4x12 screws through key/washer/B130 sleeves, nominal4.9mm engagement. Key mounting torque2.4..2.8Nm.

B130 sleeves carry clamp load; do not torque directly through unsupported polycarbonate. No adhesive retention.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



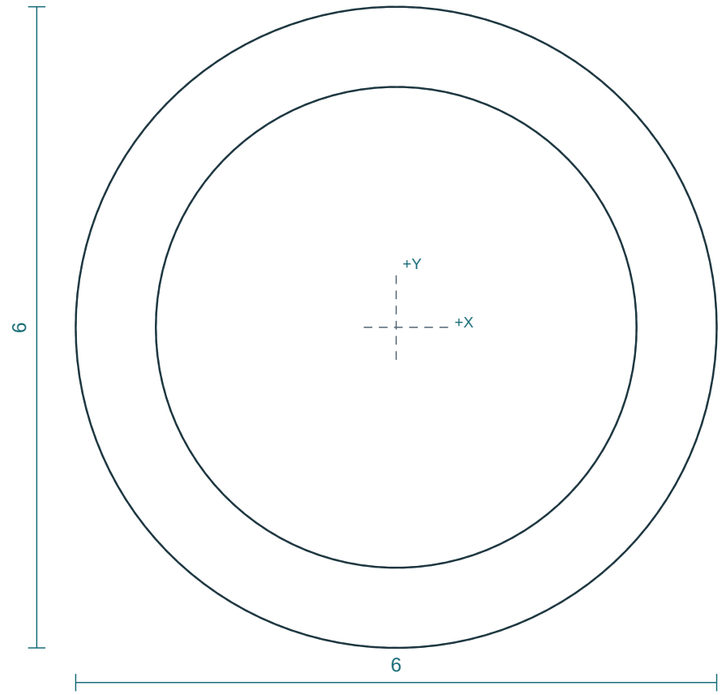
ISOMETRIC



EDGE / LOCAL XZ

Guard-key compression sleeve

Qty 2 | 303 stainless | Local part coordinates; see assembly placement



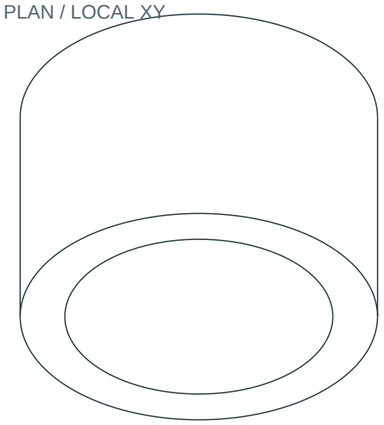
OD6, ID4.5, nominalL4.1. Finish actual length to measured panel thickness +0.10/+0.15; record finished length.

Faces parallel0.02, deburr0.1. Two sleeves in D7 panel holes. Sleeve ends bear on key and B129, not plastic.

Changing sleeve length changes key insertion datum; verify40..42.5mm per S23 after clamping.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



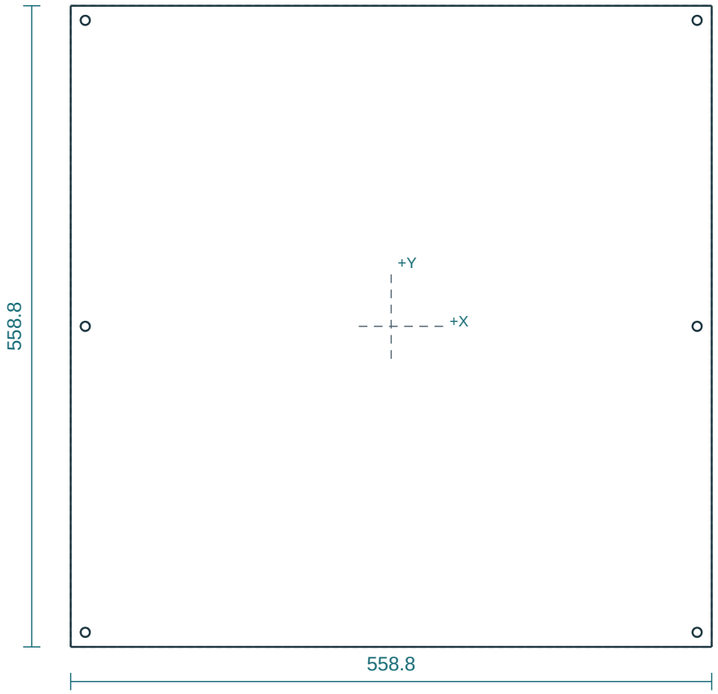
ISOMETRIC



EDGE / LOCAL XZ

Side guard blank

Qty 2 | Clear polycarbonate | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	-266.7	-266.7	D8 THRU; floating panel fastener
H2	-266.7	0	D8 THRU; floating panel fastener
H3	-266.7	266.7	D8 THRU; floating panel fastener
H4	266.7	-266.7	D8 THRU; floating panel fastener
H5	266.7	0	D8 THRU; floating panel fastener
H6	266.7	266.7	D8 THRU; floating panel fastener

Blank 558.8 x 558.8 x 4.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

Saw / router blank; deburr R0.5 edges.

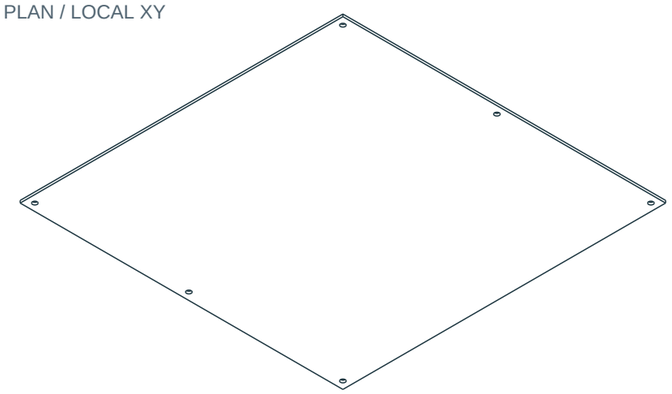
3393 1/4-20 x 1/2 screws + 18 OD x 6.6 ID x 1 mm washers; 2 N m initial assembly torque.

B128-B130 define interlock/key mounts. Physical insertion/access test remains P3. No hinge fitted.

Do not substitute acrylic for polycarbonate.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY

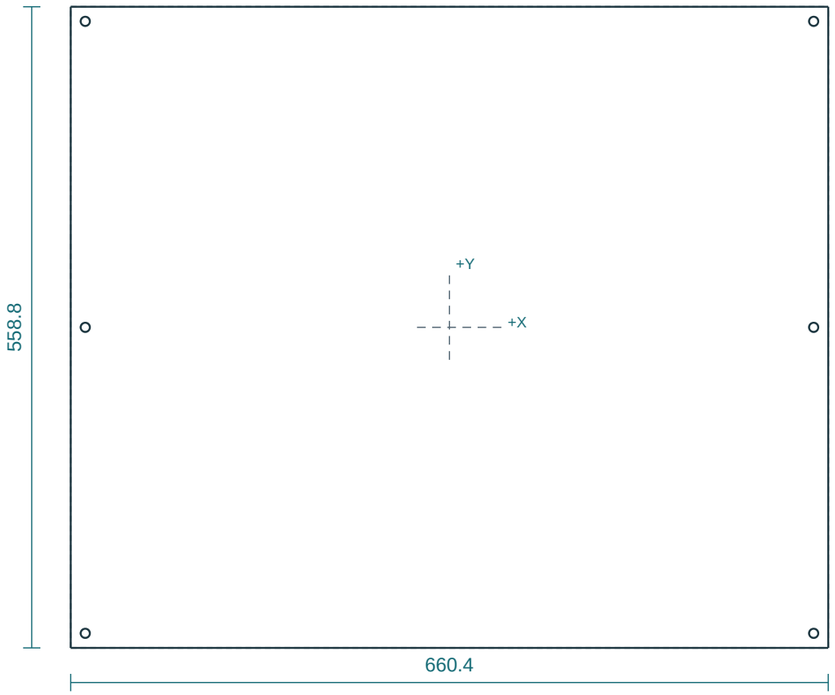


ISOMETRIC

EDGE / LOCAL XZ

Rear guard blank

Qty 1 | Clear polycarbonate | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	-317.5	-266.7	D8 THRU; floating panel fastener
H2	-317.5	0	D8 THRU; floating panel fastener
H3	-317.5	266.7	D8 THRU; floating panel fastener
H4	317.5	-266.7	D8 THRU; floating panel fastener
H5	317.5	0	D8 THRU; floating panel fastener
H6	317.5	266.7	D8 THRU; floating panel fastener

Blank 660.4 x 558.8 x 4.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

Saw / router blank; deburr R0.5 edges.

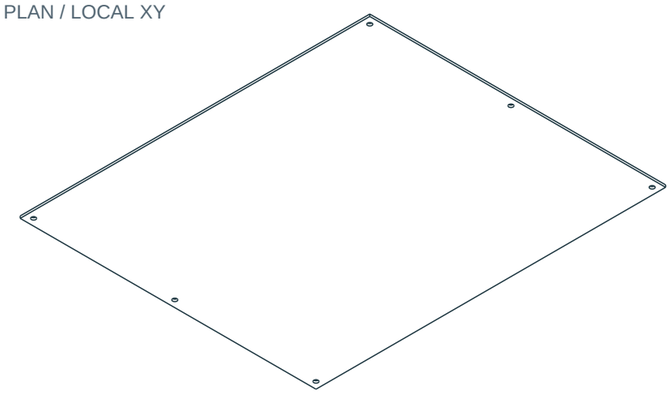
3393 1/4-20 x 1/2 screws + 18 OD x 6.6 ID x 1 mm washers; 2 N m initial assembly torque.

B128-B130 define interlock/key mounts. Physical insertion/access test remains P3. No hinge fitted.

Do not substitute acrylic for polycarbonate.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY

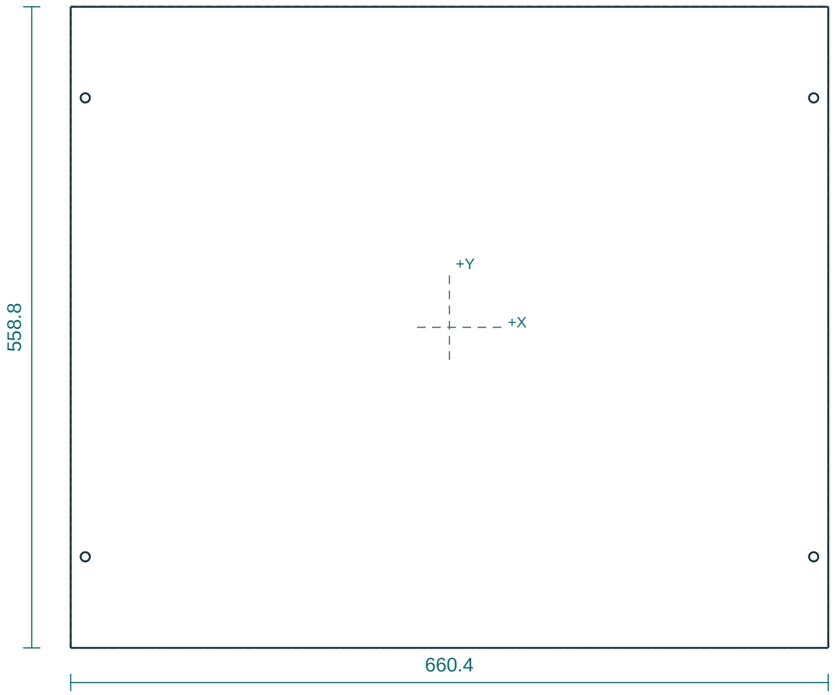


ISOMETRIC

EDGE / LOCAL XZ

Roof guard blank

Qty 1 | Clear polycarbonate | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	-317.5	-200	D8 THRU; floating panel fastener
H2	-317.5	200	D8 THRU; floating panel fastener
H3	317.5	-200	D8 THRU; floating panel fastener
H4	317.5	200	D8 THRU; floating panel fastener

Blank 660.4 x 558.8 x 4.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

Saw / router blank; deburr R0.5 edges.

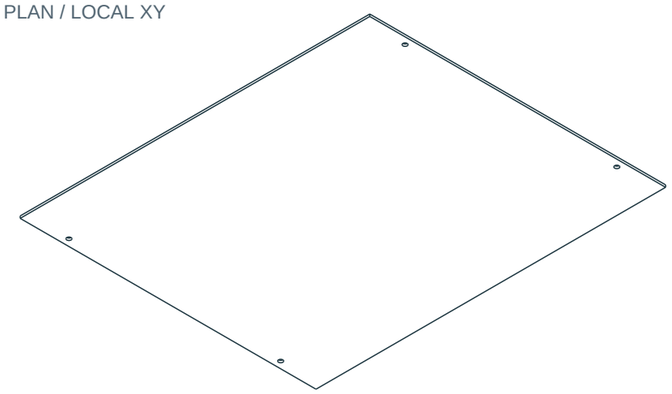
3393 1/4-20 x 1/2 screws + 18 OD x 6.6 ID x 1 mm washers; 2 N m initial assembly torque.

B128-B130 define interlock/key mounts. Physical insertion/access test remains P3. No hinge fitted.

Do not substitute acrylic for polycarbonate.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY

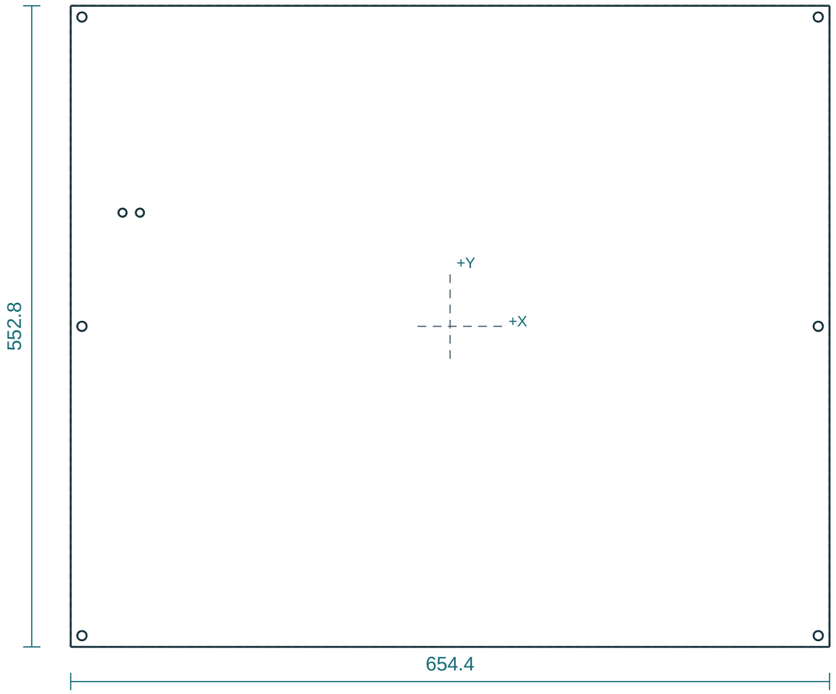


ISOMETRIC

EDGE / LOCAL XZ

Front access panel

Qty 1 | Clear polycarbonate | Local part coordinates; see assembly placement



ID	X	Y	Feature
H1	-317.5	-266.7	D8 THRU; floating panel fastener
H2	-317.5	0	D8 THRU; floating panel fastener
H3	-317.5	266.7	D8 THRU; floating panel fastener
H4	317.5	-266.7	D8 THRU; floating panel fastener
H5	317.5	0	D8 THRU; floating panel fastener
H6	317.5	266.7	D8 THRU; floating panel fastener
H7	-282.5	98	D7 THRU; key sleeve B130
H8	-267.5	98	D7 THRU; key sleeve B130

Blank 654.4 x 552.8 x 4.

Datum: XY center; Z=0 bottom. All holes THRU unless a thread or depth is stated.

Saw / router blank; deburr R0.5 edges.

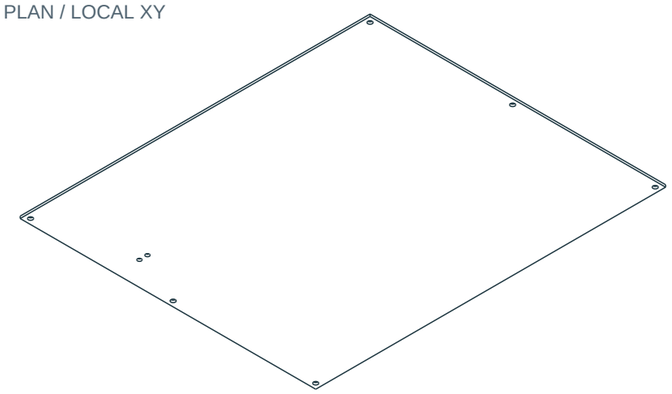
3393 1/4-20 x 1/2 screws + 18 OD x 6.6 ID x 1 mm washers; 2 N m initial assembly torque.

B128-B130 define interlock/key mounts. Physical insertion/access test remains P3. No hinge fitted.

Do not substitute acrylic for polycarbonate.

General: machined dimensions +/-0.10; angles +/-0.5 deg unless stated. Deburr 0.2-0.5. Threads are modeled at tap-drill diameter; apply callouts. Do not infer thread size from STEP.

PLAN / LOCAL XY



ISOMETRIC

EDGE / LOCAL XZ

Procurement and fabrication register

No generic fastener kits | quantities exclude spares | RFQ is not an orderable confirmed SKU

ID	Qty	Manufacturer / part	Specification / disposition	Evidence
V01	4	80/20 1010	Cut660.4 mm; front/rear rails	S01 / SELECTED
V02	8	80/20 1010	Cut508.0 mm; four side rails/four posts	S01 / SELECTED
V03	2	80/20 1010	Cut609.6 mm; base support crossmembers	S01 / SELECTED
V04	28	80/20 4119	Inside corner bracket; see FR-001 schedule	S02 / SELECTED
V05	90	80/20 3393	56 structure +28 guard panels +4 base +2 interlock mount. Each set is screw and T-nut; no metric substitution.	S03 / SELECTED
V06	32	Washer to drawing F-W14	18 OD x6.6 ID x1 mm stainless; 28 panels +4 base recesses	SHOP SPEC / SELECTED
V08	1	ROBOTIS XM430-W350-T	TTL version; use included horn; verify horn/case screw lengths against S04	S04 / SELECTED
V09	2	NSK 626	Open bearing6x19x6; protected clean interior	S07 / SELECTED
V10	4	Ruland MCL-6-A	6 mm bore one-piece clamp collar; supplied screw	S08 / SELECTED
V11	2	HIWIN MGN7C + MGNR7,100 mm	One block per100mm rail; 7 mounting holes,5mm end distance,zero/light preload. Confirm order code, block thread pitch and seal length before purchase.	S06 / RFQ
V12	1	Dynatect EAS12,6mm/6mm	Configured shaft-to-shaft, adjustable; request0.20 N m bidirectional setpoint. Designation is RFQ wording, not a confirmed order code.	S05 / HOLD P2

Procurement and fabrication register

No generic fastener kits | quantities exclude spares | RFQ is not an orderable confirmed SKU

ID	Qty	Manufacturer / part	Specification / disposition	Evidence
V20	1	Hammond 1554YA2GY	Polycarbonate box/lid/gasket/six screws; independent benchtop cabinet	S13 / SELECTED
V21	1	Hammond 1554YPL	Factory internal steel panel; mounting screw set1591MM100 as required by purchased enclosure	S13 / SELECTED
V22	1	MEAN WELL GST60A12-P1J	External12V5A supply; region-matched factory AC lead; DC inlet adapter still requires rated cable assembly	S14 / SELECTED
V23	1	MEAN WELL GST25A24-P1J	External24V1.04A control supply; region-matched factory AC lead	S15 / SELECTED
V24	1	Neutrik NC3FD-LX	12V DC panel inlet; mating NC3MXX. Project pin1=0V,2=+12V,3 unused. Label DC ONLY.	S16/S17 / SELECTED
V25	1	Neutrik NC4FD-LX	24V DC panel inlet; mating NC4MXX. Project pin1=0V,4=+24V,2/3 unused. Label DC ONLY.	S18/S19 / SELECTED
V26	2	Pilz PNOZ X3 774310	K1 emergency stop; K2 guard. Separate relays avoid series input fault masking; wiring review required.	S20 / HOLD P4
V27	1	Schneider XB5AS8444	2NC twist-release emergency stop; add yellow background and anti-rotation hardware per manufacturer	S21 / SELECTED
V28	2	Schneider XB5AA61	Separate blue reset buttons, one per safety relay; label E-STOP RESET and GUARD RESET	S22 / SELECTED
V29	1	OMRON D4NS-1BF + D4DS-K2	Guard switch/key. PG13.5 plastic gland required. Mounts B128-B130; physical insertion/access acceptance required	S23 / HOLD P3
V30	20	Phoenix UT 2,5 /3044076	Terminal blocks; E-004 defines X1:1..20 and bus bridges. Wire sizing/routing and physical release remain P4	S25 / HOLD P4

Procurement and fabrication register

No generic fastener kits | quantities exclude spares | RFQ is not an orderable confirmed SKU

ID	Qty	Manufacturer / part	Specification / disposition	Evidence
V31	2	OMRON D2F-01L-D3	Endpoint low-load switch with solder terminals; B125-B127 mount/cam; verify actual overtravel and signal timing	S24 / HOLD P3
V32	1	Raspberry Pi 5,4GB	Outside sealed control enclosure, with official active cooler and Pi5 case; use physical-board fit check	S10/S11 / SELECTED
V33	1	ROBOTIS U2D2	USB to TTL interface; mount on hub with supplied rivet	S26 / SELECTED
V34	1	ROBOTIS U2D2 Power Hub Board Set	48x57; supplied M3 supports and100mm TTL lead. Power from screw terminals, not2.5mm DC jack.	S26 / SELECTED
V35	1	ROBOTIS Robot Cable-X3P,240mm	Confirm available factory length and route; no spliced TTL cable through moving pinch points	S12/S26 / RFQ
V36	2	Analog Devices DS18B20+,TO-92	Powered3-wire sensors; case probe attachment and insulated housing to be defined	S27 / HOLD P4
V37	1	Raspberry Pi27W USB-C supply	External official supply, regional plug. Independent Pi power maintains fault logging.	S10 / SELECTED / verify accessories
V38	4	Phoenix 3022276 / CLIPFIX35-5	Two end brackets per DIN rail; width5.2mm	S30 / SELECTED / P4 fit review
V39	1	Phoenix 3047028 / D-UT2,5/10	End cover at open side of the20-terminal bank	S31 / SELECTED / P4 fit review
V40	1	Phoenix 0801733 / NS35/7.5 PERF2000MM	Cut2x140mm;280mm used before saw waste	S32 / SELECTED / P4 fit review
V41	2	Phoenix 3030187 / FBS4-5	Bridge X1:5..8(+24V) and9..12(0V24) only	S33 / SELECTED / P4 fit review

Procurement and fabrication register

No generic fastener kits | quantities exclude spares | RFQ is not an orderable confirmed SKU

ID	Qty	Manufacturer / part	Specification / disposition	Evidence
V42	2	Littelfuse FHAC0001ZXJ	Covered32VDC20A inline holder; F1/F2=V52/V53; P4 coordination	S35 / SELECTED / P4 fit review
F01	22	ISO4762 M2 x6	14 rail +8 carriage screws; block pitch confirmation required; project torque 0.15 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F02	8	ISO4762 M2 x12	Four travel stops;4mm engagement after8mm POM; project torque 0.10 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F03	8	ISO4762 M2.5 x8	Two motor cheeks;2mm case engagement; project torque 0.20 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F04	4	ISO4762 M2 x8	Motor flange6mm +0.2 shim washer;1.8mm horn engagement; project torque 0.10 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F05	8	ISO4762 M3 x8	Bearing retaining plates;6mm thread engagement; project torque 0.5 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F06	8	ISO4762 M3 x12	Yoke arms6mm to hub flange6mm; project torque 0.5 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F07	8	ISO4762 M4 x10	Four crossbar end joints +four guide carrier joints; project torque 1.0 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F08	8	ISO4762 M4 x16	Four counterweight screws +four motor cheek/foot screws; project torque 1.5 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F09	4	ISO4762 M4 x20	Counterweight adjustment plates; washer/nyloc nut stack; project torque 1.5 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F10	8	ISO4762 M4 x12	Four fixed tray screws(6mm engagement) +four endpoint upright feet with1mm washers(5mm engagement); project torque 1.0 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending

Procurement and fabrication register

No generic fastener kits | quantities exclude spares | RFQ is not an orderable confirmed SKU

ID	Qty	Manufacturer / part	Specification / disposition	Evidence
F11	8	ISO4762 M5 x12	Tower to angle;6mm engagement in8mm tower; project torque 2.0 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F12	4	ISO4762 M5 x14	Angle6mm to sole10mm;8mm engagement; project torque 2.0 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F13	10	ISO4762 M5 x16	Four tray posts +two stop posts +four motor-foot slots; project torque 2.0 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F14	4	ISO4762 M6 x16	Tower soles;recessed2mm and1mm washer =>7mm engagement; project torque 3.0 initial N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F15	2	ISO4762 M6 x25	Angular stop screws;M6 jam nuts;minimum10mm engagement; project torque lock after setup N m (not a vendor rating). Clean dry fasteners; do not apply generic steel torque to aluminum threads.	S28 / SPECIFIED / vendor SKU pending
F16	6	ISO4026 M3x4 flat-point set screw	Two per output hub +two cam; shaft flats required; proof test0.75 N m	SHOP SPEC / SELECTED
F17	4	M4 prevailing-torque locknut	B120 to B106 joints; replace after repeated adjustment if retention declines	SHOP SPEC / SELECTED
F18	2	M6 thin jam nut	Angular-stop screw locking	SHOP SPEC / SELECTED
F19	16	M4 washer,1mm	Four weight screws +eight adjustment faces +four endpoint feet	SHOP SPEC / SELECTED
F20	8	M5 washer,1mm	Four motor slots +four endpoint sole mounts	SHOP SPEC / SELECTED
F21	4	M6 washer,1mm	Recessed tower sole bolts	SHOP SPEC / SELECTED

Procurement and fabrication register

No generic fastener kits | quantities exclude spares | RFQ is not an orderable confirmed SKU

ID	Qty	Manufacturer / part	Specification / disposition	Evidence
F22	4	D2.2 ID x5 OD x0.2 shim washer	Horn screws; inspect thickness	SHOP SPEC / SELECTED
F23	4	ISO4762 M2x16	Endpoint switch through-bolts	S24 / SHOP SPEC / SELECTED
F24	4	ISO7089 M2 plain washer	Endpoint switch screw-head seats; nominal0.3mm	S24 / SHOP SPEC / SELECTED
F25	4	ISO4032 M2 hex nut	Endpoint switch rear nuts	S24 / SHOP SPEC / SELECTED
F26	4	DIN127 M2 spring washer	Between switch upright and nut; compressed stack checked	S24 / SHOP SPEC / SELECTED
F27	2	ISO4026 M3x4 flat-point set screw	Acetal endpoint cam; initial0.10Nm; NOT the load-bearing cam	S24 / SHOP SPEC / SELECTED
F28	4	ISO4762 M5x20	Endpoint sole12mm +washer1mm;7mm thread engagement in base	S24 / SHOP SPEC / SELECTED
F29	2	ISO4762 M4x12	D4DS-K2 key; washer/sleeve stack; torque2.4..2.8Nm	S24 / SHOP SPEC / SELECTED
F30	2	ISO4762 M4x50 blank	D4NS body: cut to actual stack +nut +two full threads; record final length. Body clamp0.5..0.7Nm	S24 / SHOP SPEC / SELECTED
F31	6	ISO7089 M4 washer,1mm	Two key heads +four switch body through-bolt faces	S24 / SHOP SPEC / SELECTED
F32	2	ISO4032 M4 hex nut	Guard-switch body rear nuts	S24 / SHOP SPEC / SELECTED

Procurement and fabrication register

No generic fastener kits | quantities exclude spares | RFQ is not an orderable confirmed SKU

ID	Qty	Manufacturer / part	Specification / disposition	Evidence
F33	2	DIN127 M4 spring washer	Guard-switch body nut seats	S24 / SHOP SPEC / SELECTED
C01	200x200mm	Silicone sheet1mm,30-40 Shore A	Contact pads and stop-cap pads; compatibility with phone coating to be checked	SHOP SPEC / HOLD P1
C02	100x100mm	Silicone sheet2mm,30-40 Shore A	Travel-stop pads; 4 pieces6x8; measure actual float after bonding	SHOP SPEC / SELECTED
C03	4 pieces	Neoprene pad25x50x3	Under bottom frame, near four corners; compression/rocking checked on bench	SHOP SPEC / SELECTED
V43	1	Adafruit1609	Controller base PCB; E-005..008	S36 / SELECTED / P4 acceptance
V44	1	LITEON LTV-817-C DIP4	U1; CTR rankC	S37 / SELECTED / P4 acceptance
V45	1	Vishay1N4148	D1 reverse-LED protection	S39 / SELECTED / P4 acceptance
V46	1	Vishay PR01 2kohm 1% 1W	R1 permit input	S38 / SELECTED / P4 acceptance
V47	1	Vishay PR01 10kohm 1% 1W	R2 pulldown	S38 / SELECTED / P4 acceptance
V48	1	Vishay PR01 1kohm 1% 1W	R3 GPIO series	S38 / SELECTED / P4 acceptance
V49	2	Vishay PR01 1.8kohm 1% 1W	R4,R6	S38 / SELECTED / P4 acceptance

Procurement and fabrication register

No generic fastener kits | quantities exclude spares | RFQ is not an orderable confirmed SKU

ID	Qty	Manufacturer / part	Specification / disposition	Evidence
V50	2	Vishay PR01 2.7kohm 1% 1W	R5,R7	S38 / SELECTED / P4 acceptance
V51	1	Vishay PR01 4.7kohm 1% 1W	R8 1Wire	S38 / SELECTED / P4 acceptance
V52	1	Littelfuse0287002.L	F1 2A32VDC; holderV42	S40 / SELECTED / P4 acceptance
V53	1	Littelfuse0287001.L	F2 1A32VDC; holderV42	S40 / SELECTED / P4 acceptance
V54	1	LAPP53015130	Guard PG13.5 plastic gland	S41 / SELECTED / P4 acceptance
V55	1.8m	LAPP1119754	Guard4x0.5 cable; trim	S42 / SELECTED / P4 acceptance
V56	1	Hammond1427BCGM16	Guard cabinet entry only	S34 / SELECTED / P4 acceptance
E-HOLD	1 lot	Electrical completion schedule	Fuse coordination acceptance; wire/ferrules/glands; board mounting/cover; DC adapter cables; cable anchors; Pi breakout. These are not procurement-ready generic kits: release P4 first.	UNRESOLVED / HOLD P4
DUT-HOLD	1 set	Phone inserts and dummy	Shop measures actual phone before machining inserts, as authorized by owner. No invented Apple dimensions.	SHOP MEASUREMENT / HOLD P1
B101	1	Custom drawing B101	Fixture base; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B102	2	Custom drawing B102	Bearing tower; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW

Procurement and fabrication register

No generic fastener kits | quantities exclude spares | RFQ is not an orderable confirmed SKU

ID	Qty	Manufacturer / part	Specification / disposition	Evidence
B103	2	Custom drawing B103	Tower sole plate; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B104	2	Custom drawing B104	Tower angle; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B105	2	Custom drawing B105	Bearing retaining plate; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B106	2	Custom drawing B106	Yoke / counterbalance arm; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B107	1	Custom drawing B107	Lightened crossbar; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B108	2	Custom drawing B108	Guide carrier strip; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B109	1	Custom drawing B109	Floating contact plate blank; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B110	4	Custom drawing B110	Guide travel stop; POM-C acetal; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B111	2	Custom drawing B111	Counterweight; Low-carbon steel; zinc plate after machining; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B112	2	Custom drawing B112	Output flange hub; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B113	1	Custom drawing B113	Ground output shaft; Ground stainless steel, D6 g6; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW

Procurement and fabrication register

No generic fastener kits | quantities exclude spares | RFQ is not an orderable confirmed SKU

ID	Qty	Manufacturer / part	Specification / disposition	Evidence
B114	1	Custom drawing B114	Ground output shaft; Ground stainless steel, D6 g6; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B115	2	Custom drawing B115	Inner-race spacer; 303 stainless; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B116	2	Custom drawing B116	Inner-race spacer; 303 stainless; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B117	1	Custom drawing B117	Fixed tray; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B118	4	Custom drawing B118	Fixed tray standoff; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B119	2	Custom drawing B119	Motor mounting side cheek; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B120	2	Custom drawing B120	Counterweight adjustment plate; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B121	1	Custom drawing B121	Motor adjustable pedestal; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B122	1	Custom drawing B122	Integral motor input flange / shaft; 303 stainless steel; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B123	1	Custom drawing B123	Output angular-stop cam; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B124	2	Custom drawing B124	Angular-stop post; 303 stainless steel; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW

Procurement and fabrication register

No generic fastener kits | quantities exclude spares | RFQ is not an orderable confirmed SKU

ID	Qty	Manufacturer / part	Specification / disposition	Evidence
B125	1	Custom drawing B125	Endpoint sensing cam; POM-C acetal; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B126	1	Custom drawing B126	Endpoint stand sole; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B127	2	Custom drawing B127	Endpoint switch upright; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B128	1	Custom drawing B128	Guard-switch post bracket; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B129	1	Custom drawing B129	Guard-key backing plate; 6061-T6 aluminum; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B130	2	Custom drawing B130	Guard-key compression sleeve; 303 stainless; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B201	2	Custom drawing B201	Side guard blank; Clear polycarbonate; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B202	1	Custom drawing B202	Rear guard blank; Clear polycarbonate; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B203	1	Custom drawing B203	Roof guard blank; Clear polycarbonate; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW
B204	1	Custom drawing B204	Front access panel; Clear polycarbonate; individual STEP supplied	DRAWING BOOK / DEVELOPMENT REVIEW

Vendor evidence and CAD provenance

Access checked 2026-11-05 | downloaded files are hashed in source-register.json

S01 / 80/20 / 1010 extrusion
25.4 square; current product lists 6063-T6; I=0.0442 in^4; weight 0.0424 lb/in.
Reference URL only; no local source file; Reconstructed envelope; slot relief is illustrative. Original STEP requires PartCommunity sign-in.
https://8020.net/1010.html

S02 / 80/20 / 4119 inside corner bracket
25.4 legs; 22.225 width; 4.7752 thick; D7.1374 holes at 12.7.
Reference URL only; no local source file; Reconstructed mounting geometry; edge radii omitted.
https://8020.net/4119.html

S03 / 80/20 / 3393 button-head screw / T-nut set
1/4-20 x 0.500 screw; 10-series slide-in economy T-nut. Two sets per 4119 bracket.
Reference URL only; no local source file; No original CAD downloaded; do not manufacture T-nuts from the assembly illustration.
https://8020.net/3393.html

S04 / ROBOTIS / XM430-W350-T/R manufacturer drawing
28.5 x 46.5 x 34; axis 11.25 below top; side pattern 12 x 24; horn PCD16. M2 horn max 2 deep; M2.5 case max 3 deep.
Downloaded; signature verified; Reconstructed interface model. Manufacturer's legacy STEP links returned HTML, discarded.
https://media.distrelec.com/Web/Downloads/_t/ds/Dynamixel_XM-Series_eng_tds.pdf

S05 / Dynatect / Polyclutch Slip-Ease catalogue
EAS12: OD19.05, overall31.75, bossOD14.28, bossL4.78. Published max0.96 N m at50rpm; 4.5W. Exact 6mm bores require configuration.
Downloaded; signature verified; Envelope only; insertion depth and low-torque setting not confirmed.
https://www.dynatect.de/wp-content/uploads/Polyclutch-SlipEaseClutch.pdf

S06 / HIWIN / Linear Guideway G99TJ22-2503, printed pp86/90
MGN7C H8 W17 L22.5 Lmax23.1; block pattern12x8, M2x2.5 deep; rail7x4.8, pitch15, end5, D2.4/D4.2 counterbore2.3. Torque table says M2x0.25 while dimension table omits pitch: confirm thread on order.
Downloaded; signature verified; Reconstructed interface, no bearing-ball or raceway manufacturing geometry.
https://www.hiwinsupport.com/download/tech_doc/gw/Linear_Guideway-%28J%29.pdf

S07 / NSK / Bearing and Linear Replacement Guide
626 open deep-groove bearing: 6 ID x19 OD x6 width; shaft shoulder8, housing shoulder17, corner radius max0.3.
Downloaded; signature verified; Reconstructed ring envelope. Buy genuine 626; do not make a bearing from STEP.
https://www.nsk.com/content/dam/nsk/am/en_us/documents/bearings-americas/Bearing-and-Linear-Replacement-Guide.pdf

Vendor evidence and CAD provenance

Access checked 2026-11-05 | downloaded files are hashed in source-register.json

S08 / Ruland / MCL-6-A one-piece clamp collar
Bore6, OD16, width9; M3x8 screw; 2.5mm hex; seating torque2.1 N m. Clamp clearance OD20.8.
Reference URL only; no local source file; Reconstructed fit envelope; clamp screw clearance must be checked on purchased collar.
https://www.ruland.com/mcl-6-a.html

S09 / ACE / FRT-C2-201 damper operating instructions
0.020 N m +/-0.006 at20rpm/23C; bodyOD15; D4 shaft flat3.5; mount centers21; 50,000 reference cycles. No shaft side/axial load.
Downloaded; signature verified; Investigated, not fitted: torque/lifetime do not justify the previous generic damper assumption.
https://www.acecontrols.co.uk/media/msimages/pdf/ACE_FRT%28FRN%29-C2_Operating-Mounting_EN_21_22_0019.pdf

S10 / Raspberry Pi / Pi 5 mechanical drawing
Board85x56; mounting pitch58x49; D2.7 mounting holes. Vendor explicitly calls dimensions approximate and not production data.
Downloaded; signature verified; Use physical board before machining a close-fitting enclosure.
https://datasheets.raspberrypi.com/rpi5/raspberry-pi-5-mechanical-drawing.pdf

S11 / Raspberry Pi / Pi 5 original STEP archive
Original manufacturer 3D board/component model; archive includes LICENSE.txt (MIT).
Downloaded; signature verified; Original vendor STEP; preserve included license.
https://pip-assets.raspberrypi.com/categories/892-raspberry-pi-5/documents/RP-010083-CA-1-rpi-5%203D%20STEP%20-%20No%20Graphics%20small%20file.zip

S12 / JST / EH connector family
2.5mm pitch EH family. ROBOTIS TTL uses EHR-03 housing, B3B-EH-A header; 1 GND / 2 VDD / 3 DATA.
Downloaded; signature verified; Factory Robot Cable-X3P preferred; no generic Dupont substitutes.
https://www.jst-mfg.com/product/pdf/eng/eEH.pdf

S13 / Hammond / 1554YA enclosure drawing
1554YA2GY polycarbonate variant:299.33x239.46x119.73; inside height111.75. Optional1554YPL panel250.50x214.52. Lid torque0.9-1.0 N m.
Downloaded; signature verified; Reconstructed shell envelope; bosses and seals omitted. STEP not obtained.
https://www.hammfg.com/files/parts/pdf/1554YAGY.pdf

S14 / MEAN WELL / GST60A supply series
GST60A12-P1J:12V/5A; nominal125x50x31.5 housing; P1J DC plug5.5OD/2.1ID, center positive.
Downloaded; signature verified; External factory mains brick; do not put mains terminals inside bench.
https://www.meanwell.com/Upload/PDF/GST60A/GST60A-SPEC.PDF

Vendor evidence and CAD provenance

Access checked 2026-11-05 | downloaded files are hashed in source-register.json

S15 / MEAN WELL / GST25A supply series
GST25A24-P1J:24V/1.04A; external adapter; P1J center-positive5.5OD/2.1ID plug. Use the series drawing for cable exit clearance.
Downloaded; signature verified; Original datasheet; no 3D body model needed inside cabinet.
https://www.meanwell.com/Upload/PDF/GST25A/GST25A-SPEC.PDF

S16 / Neutrik / NC3FD-LX panel connector drawing
26x31 flange; 19x24 diagonal mounting pitch; panel clearance >D23.6. Project cutout D24.0 +0.1/0; M3 mounting.
Downloaded; signature verified; Original PDF and STEP (S17); project-defined 12V pin assignment.
https://www.neutrik.com/media/8412/download/nc3fd-lx-3.pdf?v=1

S17 / Neutrik / NC3FD-LX original 3D CAD
Original vendor mechanical part geometry.
Downloaded; signature verified; Original STEP validated by file signature and OpenCascade import.
https://www.neutrik.com/media/13322/download/3-D%20NC3FD-LX.stp?v=1

S18 / Neutrik / NC4FD-LX panel connector drawing
D-series four-pin panel connector; separate keying from 3-pin 12V inlet. Same19x24 mounting pitch and D24.0 project opening.
Downloaded; signature verified; Original PDF and STEP (S19); project-defined 24V pin assignment.
https://www.neutrik.com/media/8417/download/nc4fd-lx-2.pdf?v=1

S19 / Neutrik / NC4FD-LX original 3D CAD
Original vendor mechanical part geometry.
Downloaded; signature verified; Original STEP validated by file signature and OpenCascade import. WARNING: original imports but fails overall kernel validity; retained unchanged as reference.
https://www.neutrik.com/media/11865/download/NC4FD-LX.stp?v=2

S20 / Pilz / PNOZ X3 manual 20547-EN-10
774310,24VAC/DC variant; 45 wide x121 high x87 deep. DC supply B1+/B2-. Pages12/13 define dual-channel inputs and monitored reset.
Downloaded; signature verified; Original PDF; check current manual supplied with device before wiring. Manual permits internal-use copies.
https://www.pilz.com/download/open/PNOZ_X3_Operating_Manual_20547-EN-10.pdf

S21 / Schneider Electric / XB5AS8444 emergency-stop datasheet
2NC,40mm red mushroom; D22.5 panel opening; clamp thickness1..6; 40x43x82 overall; terminal torque0.8..1.2 N m.
Downloaded; signature verified; Original manufacturer datasheet hosted by distributor.
https://www.farnell.com/datasheets/2548526.pdf

Vendor evidence and CAD provenance

Access checked 2026-11-05 | downloaded files are hashed in source-register.json

S22 / Schneider Electric / XB5AA61 reset button
Blue spring-return1NO,22mm family; panel cutout D22.5. Two separate buttons used for two safety relays.
Downloaded; signature verified; Original manufacturer datasheet; do not bridge two monitored-reset circuits.
https://media.distributordatasolutions.com/schneider_synd/2023q2/documents/772793a73ac5d61af21589d4ebbac215fb79964c.pdf

S23 / OMRON / D4NS / D4NS-SK catalogue
D4NS-1BF:2NC,PG13.5; D4DS-K2 key front-insertion setting40..42.5 from rear seating face; alignment+/-1. Mount22x47 diagonal M4; key15 pitch. Body0.5..0.7Nm; key2.4..2.8Nm. Pages7/8/10/12.
Downloaded; signature verified; Original manufacturer PDF downloaded from European server. B128-B130 project mounts; physical insertion/access tests P3.
https://files.omron.eu/downloads/latest/datasheet/en/d4ns_d4ns-sk_safety-door_switch_datasheet_en.pdf?v=1

S24 / OMRON / D2F switch datasheet
D2F-01L-D3 low-load lever,solder terminals; pitch6.5; OP6.8+/-1.5,OTmin0.55; mounting0.08..0.10Nm. Solder300C max3sec; leave undisturbed1min.
Downloaded; signature verified; Original PDF; B125-B127 define endpoint mount/cam. Actuation and wear still require physical acceptance.
https://omronfs.omron.com/en_US/ecb/products/pdf/en-d2f.pdf

S25 / Phoenix Contact / UT 2,5 terminal block 3044076
5.2 wide x47.7 long; 47.5 high on NS35/7.5; strip9mm; torque0.5..0.6 N m.
Downloaded; signature verified; Original manufacturer PDF hosted by distributor; end plates/stops must be separately ordered.
https://www.farnell.com/datasheets/3541626.pdf

S26 / ROBOTIS / U2D2 and power hub
Power-hub board48x57; 3.5..24V,10A maximum. Included M3x10 supports. Hub DC jack is2.5ID, unlike selected supply's2.1ID plug: use screw terminals.
Reference URL only; no local source file; Manufacturer web instructions; mounting pattern must be measured or vendor drawing obtained.
https://docs.robotis.com/docs/parts/interface/u2d2_power_hub/

S27 / Analog Devices / DS18B20 temperature sensor
3-wire powered1-Wire sensor; select genuine TO-92 part. A bare sensor datasheet does not define a third-party waterproof probe housing.
Reference URL only; no local source file; Primary URL identified; download failed. Probe assembly not dimensionally released.
https://www.analog.com/media/en/technical-documentation/data-sheets/ds18b20.pdf

S28 / Bossard / BN612 ISO4762 socket screw catalogue
Metric cap-head nominal envelopes d/headD/headH:2/3.8/2, 2.5/4.5/2.5,3/5.5/3,4/7/4,5/8.5/5,6/10/6.
Reference URL only; no local source file; Manufacturer catalogue indexed online; download403. Supplied STEP fasteners are standard-derived envelopes, not original vendor CAD.
https://www.tme.eu/Document/c28d0e41a5209ef4d5f4150eb50af631/BN612-IT.pdf

Vendor evidence and CAD provenance

Access checked 2026-11-05 | downloaded files are hashed in source-register.json

S29 / MISUMI / Hole fit tolerance reference
H7 for diameters over18 through30mm is +0.021/0. The D19 bearing seats are therefore19.000..19.021mm.
Reference URL only; no local source file; Manufacturer technical reference; fit callout, not a purchased part.
https://de.meviy.misumi-ec.com/help/en/technical_info-en-c/mpb_conditions-en-c/9726/

S30 / Phoenix Contact / CLIPFIX 35-5 / 3022276
Width5.2; length48.6; depth37.1 on NS35/7.5.
Reference URL only; no local source file; Manufacturer dimension table; purchased DIN end bracket.
https://www.phoenixcontact.com/en-us/products/end-block-clipfix-35-5-3022276?type=pdf

S31 / Phoenix Contact / D-UT 2,5/10 / 3047028
Width2.2; height47; depth39.8. End cover for UT terminal bank.
Reference URL only; no local source file; Manufacturer dimension table; verify open side on installation.
https://www.phoenixcontact.com/en-de/products/end-cover-d-ut-25-10-3047028?type=pdf

S32 / Phoenix Contact / NS35/7.5 PERF 2000MM / 0801733
35mm wide,7.5mm high; stock2000mm; cut two140mm lengths.
Reference URL only; no local source file; Use purchased perforated rail; deburr both cut ends.
https://www.phoenixcontact.com/en-us/products/din-rail-ns-35-75-perf-2000mm-0801733

S33 / Phoenix Contact / FBS 4-5 / 3030187
Four positions,5.2mm pitch,24A; bridge only identified buses.
Reference URL only; no local source file; Manufacturer terminal accessory; not a universal wire jumper.
https://www.phoenixcontact.com/en-us/products/jumper-fbs-4-5-3030187

S34 / Hammond / 1427BCGM16 cable gland
M16x1.5; mounting hole16; cableOD4..8; outerL21; threadL6. Locknut and EPDM ring included.
Downloaded; signature verified; Cabinet gland only. NOT a substitute for the PG13.5 entry on the guard switch; check wall and locknut stack.
https://www.hammfg.com/files/parts/pdf/1427BCGM16.pdf

S35 / Littelfuse / ATO FHAC inline fuse holder
FHAC0001ZXJ:32VDC,20A,16AWG leads with cover. Holder rating is not the selected fuse current.
Reference URL only; no local source file; Manufacturer datasheet; fuse selection/coordination remains P4.
https://www.littelfuse.com/assetdocs/littelfuse-fuse-holder-ato-fhac-datasheet.pdf?assetguid=272e0b1a-a576-4173-8740-c1eb469efd79

Vendor evidence and CAD provenance

Access checked 2026-11-05 | downloaded files are hashed in source-register.json

S36 / Adafruit / 1609 Perma-Proto original EAGLE board
81.28 x50.8; two D3.2 holes73.66 apart; grid2.54; center gap7.62.
Downloaded; signature verified; Original EAGLE CAD; copper connectivity checked; CC BY-SA3.0.
https://raw.githubusercontent.com/adafruit/Adafruit-Perma-Proto-PCB/master/adafruit%20permaproto%20halfbreadboard.brd

S37 / LITEON / LTV-817 optocoupler manufacturer datasheet
DIP4: lead pitch2.54; rows7.62; body4.6x6.5 nominal.
Downloaded; signature verified; Manufacturer PDF via distributor; pin1A,2K,3E,4C.
https://media.digikey.com/pdf/data%20sheets/lite-on%20pdfs/ltv-817_827_847%28m%2Cs%2Cs_ta1%29.pdf

S38 / Vishay / PR01 power metal-film resistor
PR01:1W at70C; bodyL6.5,D2.5; leadD0.58; select1% tolerance.
Downloaded; signature verified; Manufacturer dimensional and rating data; no vendor STEP.
https://www.vishay.com/docs/28729/pr010203.pdf

S39 / Vishay / 1N4148 switching diode
Axial glass DO35; cathode band identifies negative terminal.
Downloaded; signature verified; Manufacturer PDF; upright mounting with insulated formed lead.
https://www.vishay.com/docs/81857/1n4148.pdf

S40 / Littelfuse / ATOF287 blade fuse32VDC
19.1 wide x18.8 high x5.1; F1=0287002.L2A; F2=0287001.L1A.
Reference URL only; no local source file; Selected prototypes; verify inrush and supply fault behavior P4.
https://www.littelfuse.com/assetdocs/littelfuse_datasheet_287_atof_r2.7.pdf?assetguid=43dcdce8-8ca2-426f-8998-7e566f048d40

S41 / LAPP / SKINTOP STR reduced PG13.5 plastic gland
53015130:4..9mm cable; SW24; L41; threadL9mm.
Downloaded; signature verified; Manufacturer catalogue; matches D4NS9mm maximum thread length.
https://products.lappgroup.com/online-catalogue/cable-glands/skintop-cable-glands-plastic-pg/standard/skintop-str.html?format=pdf

S42 / LAPP / OLFLEX CLASSIC110 4X0.5
1119754:4x0.5mm2; nominalOD5.7; 300/500V; bend5xOD.
Reference URL only; no local source file; Manufacturer product data; stationary guard-switch cable.
https://www.lapp.ch/de/item/olflex-classic-110-4x05%3B1119754

Supplier confirmation / interface release

Send the relevant drawings with this sheet | record actual responses before closing P2

Interface	Required written confirmation	Evidence / result
Dynatect EAS12	Two6.000mm nominal bores; shaft-to-shaft configuration; bore tolerance, insertion depth both ends, setscrew positions, wrench access and approved shaft flat.	Drawing/revision: _____
Clutch performance	Stable bidirectional0.20Nm nominal slip; minimum adjustment range, breakaway/running variation, thermal drift, allowable slip duty and service life.	Supplier response: _____
HIWIN MGN7C	MGNR7 rail100mm;7holes at15mm pitch,5mm end distance; block length/seal code; confirm actual M2 thread pitch and maximum engagement2.5mm.	Order code: _____
ROBOTIS horn	Included horn PCD16; actual center bolt head and projection; verify B122 recess clearance>=0.5mm and1.8mm screw engagement before coupling.	Measured clearance: _____
Shaft stack	Apply the approved clutch insertion data to B113/B122; record pedestal shift, shaft end gaps and flat clocking. Confirm no axial preload.	Updated drawings: _____

Disposition: accepted / revise design / reject substitution

Mechanical owner _____ Supplier _____ Date _____

Attach supplier drawing, quotation and received-part inspection. A catalogue maximum torque is not proof of reliable operation at the much lower requested setpoint.

Verification and release boundary

Computer checks establish model consistency; physical acceptance remains separate

Check	Result	What was checked
Custom solid validity	True	OpenCascade validity for every exported custom part
Moving sweep against base, tray, posts and towers	0 overlaps	37 positions, 0..180 degrees in 5-degree steps; >0.01 mm3 threshold
Lowest sampled moving surface	6.67 mm above base	At nominal counterweight settings; excludes wiring and screw heads
Phone geometry / clutch insertion / safety validation	NOT VERIFIED	Actual phone, configured clutch drawing and assembled hardware required

P1 - Phone-contact inserts
Measure the actual iPhone Duo case, hinge-axis path, protrusions, buttons and screen keep-outs. Release contact inserts and retention features after an actual fit test. The current rectangular envelope is not verified.

P2 - Configured clutch and shaft interfaces
Obtain Dynatect's drawing for an EAS12 shaft-to-shaft clutch with two exact 6 mm bores. Confirm insertion depths, set-screw locations and stable 0.20 N m bidirectional slip. Confirm B122 center-bolt clearance and finish the shaft flats from the approved stack.

P3 - Guard sensing and endpoint sensing
Fit-check B128-B130 interlock/key mounts against the purchased devices. B125-B127 define endpoint mounts and cam. Confirm their switch actuation before hard-stop contact and guard opening before any reachable motion. Record actual switch-body bolt lengths, key sleeve lengths, insertion datum and both-channel opening behavior.

P4 - Electrical panel and machine acceptance
Finish the cabinet mounting/cutout layout, fuse coordination and wiring release for the exact purchased components. Perform the independent stop, reset, restart, cable-disconnect and thermal tests. This issue has no machine safety-category certification.

Release signature: mechanical owner _____ electrical owner _____ shop lead _____ date _____
Do not mark a hold closed merely because the associated STEP file opens or a simulated cycle passes.